Francesco Beretta
(CNRS / University of Bern / University of Neuchâtel)

The semantic web and ontologies as a means of producing well-defined and reusable research data in the HSS

University of Bern
(version 1.0, october 2026)

Francesco Beretta, 13 October 2026 CC BY-NC-SA 4.0

Not yet finished

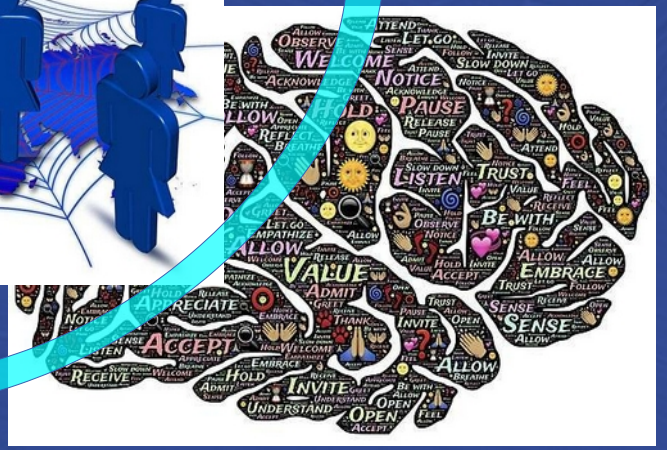
Key elements for building and managing information systems in the humanities and social sciences (HSS)

Scientific knowledge as a socially defined, complex model of a domain of interest

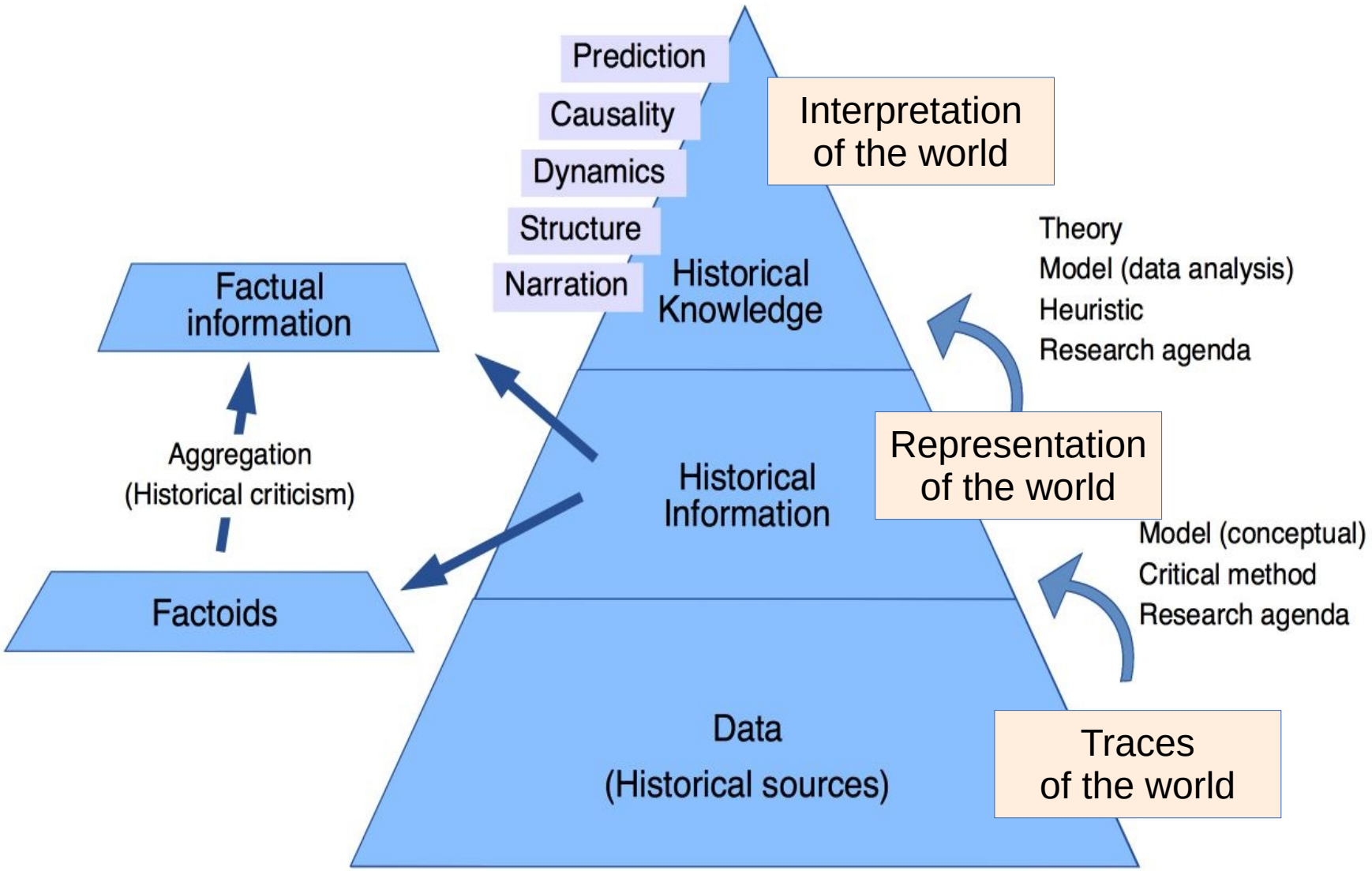








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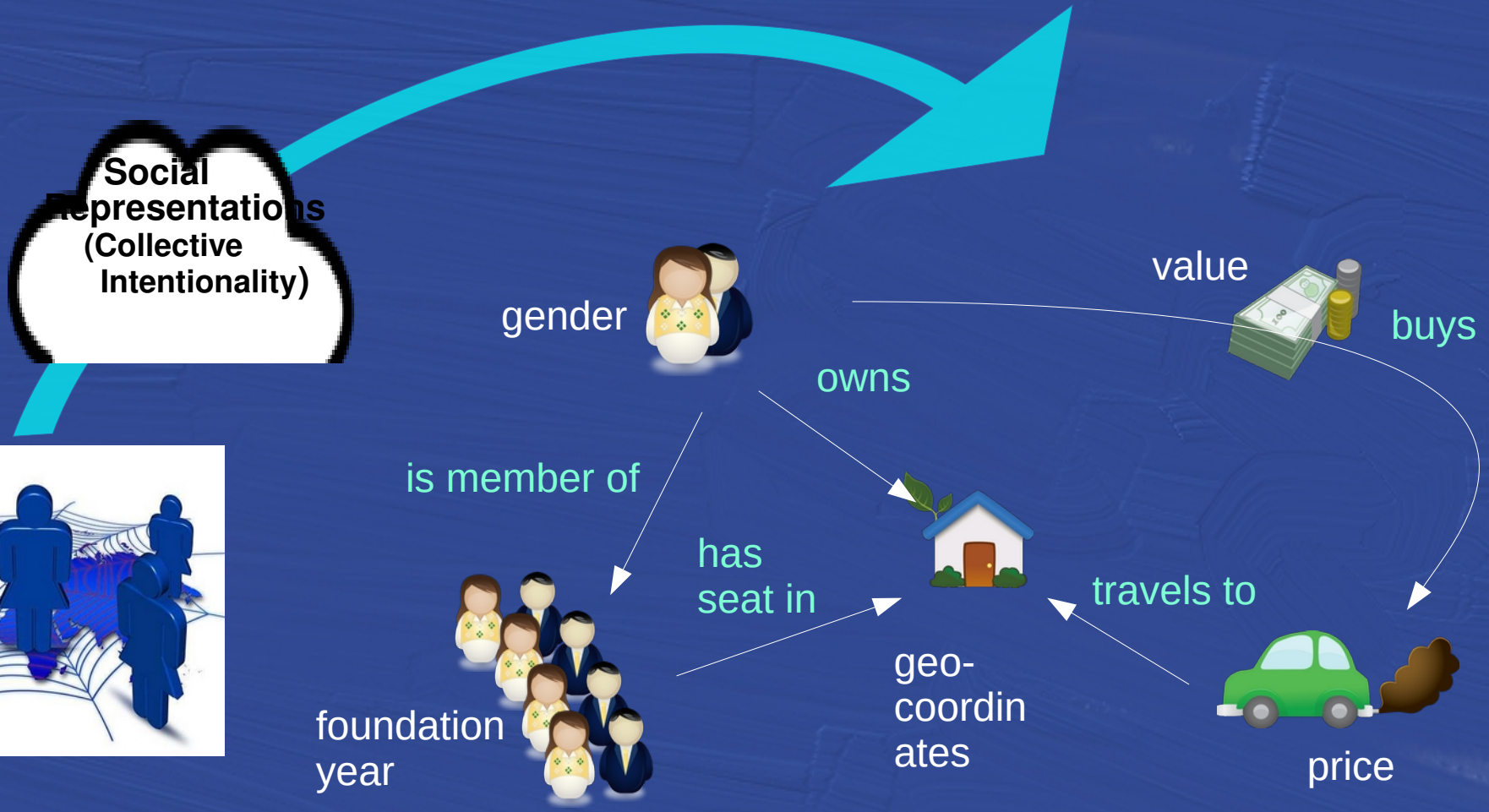


Francesco Beretta (CNRS/Université de Lyon), 7 July 2020 CC BY-NC-SA 4.0

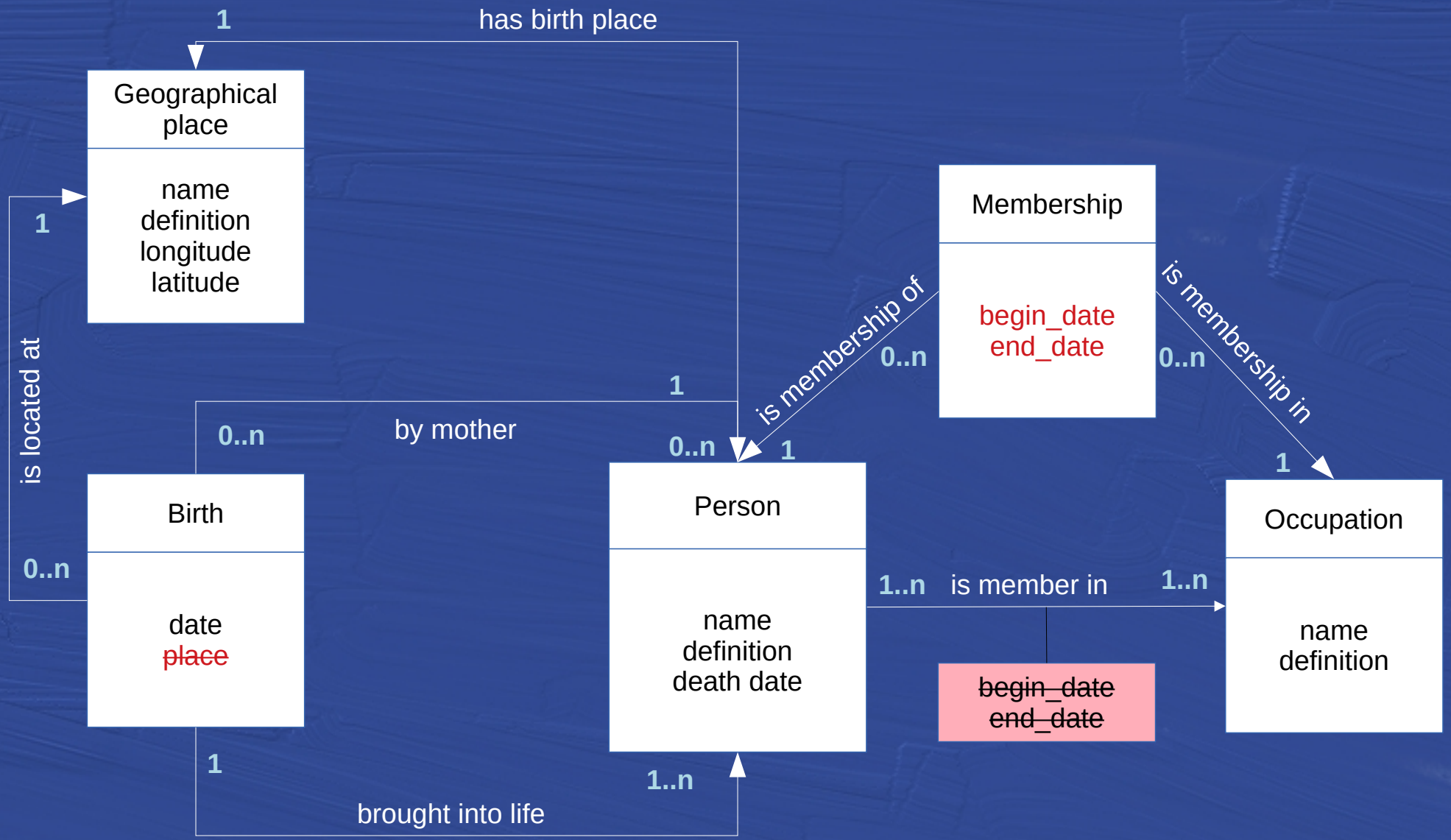
Information as **representation** of the world :

- representation of the **objects** in a domain of discourse
- of their **properties** (qualities)
- of their **relationships**

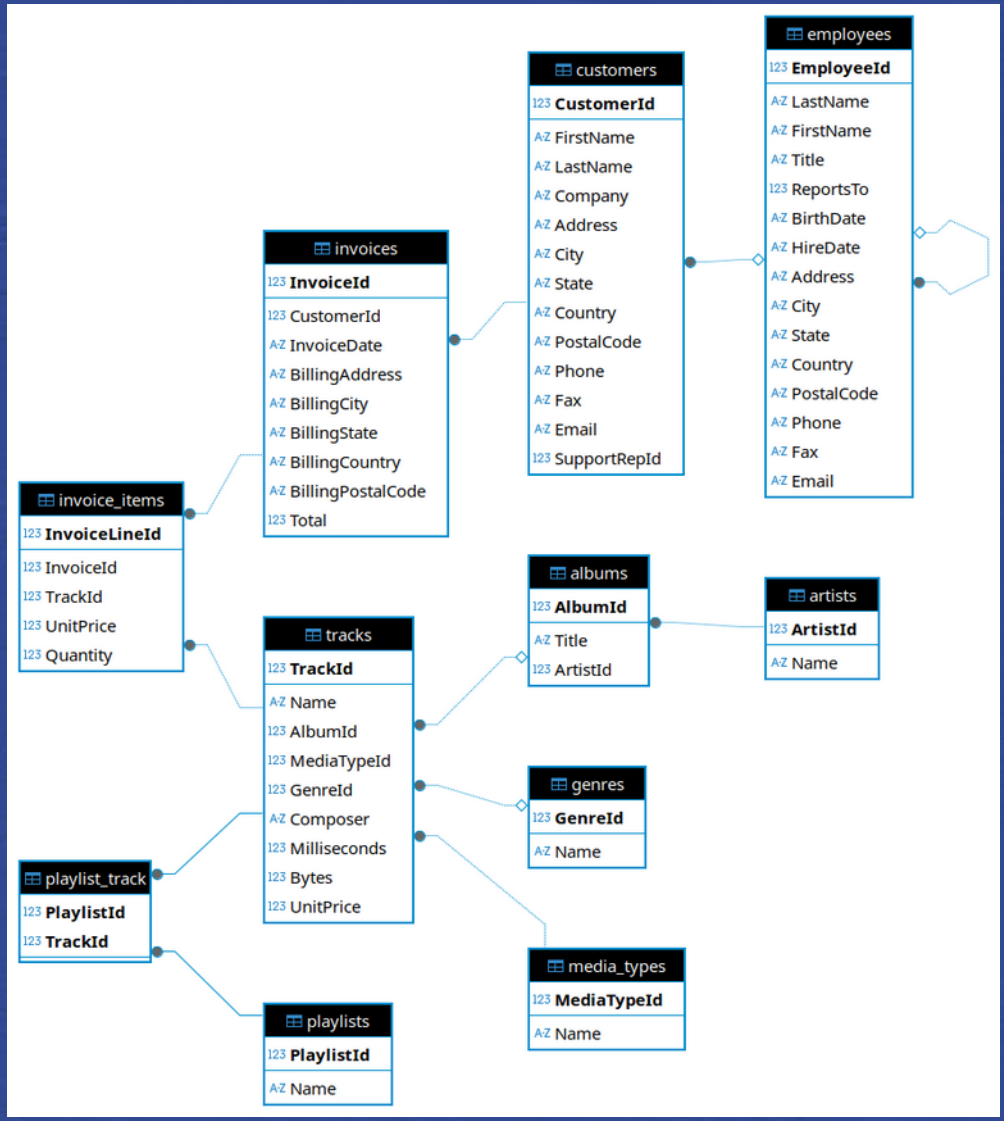




Conceptualizing the own research domain (domain of scientific discourse)



Relational databases as well established, robust technology for building information systems



Linked Open Data and the Semantic Web



Tim Berners-Lee, the inventor of the Web and Linked Data initiator, suggested a 5-star deployment scheme for Open Data.

https://5stardata.info/en/



★ make your stuff available **on the Web** (whatever format) under an open licence

★★ make it available as **structured data** (e.g., Excel instead of image scan of a table)

★★★ make it available in a non-proprietary **open format** (e.g., CSV instead of Excel)

★★★★ use **URIs to denote things**, so that people can point at your stuff

★★★★★ **link your data** to other data to provide context

Quoted from : https://5stardata.info/en/

Linked Open Data and the Semantic Web

(https://www.w3.org/TR/rdf11-concepts/)

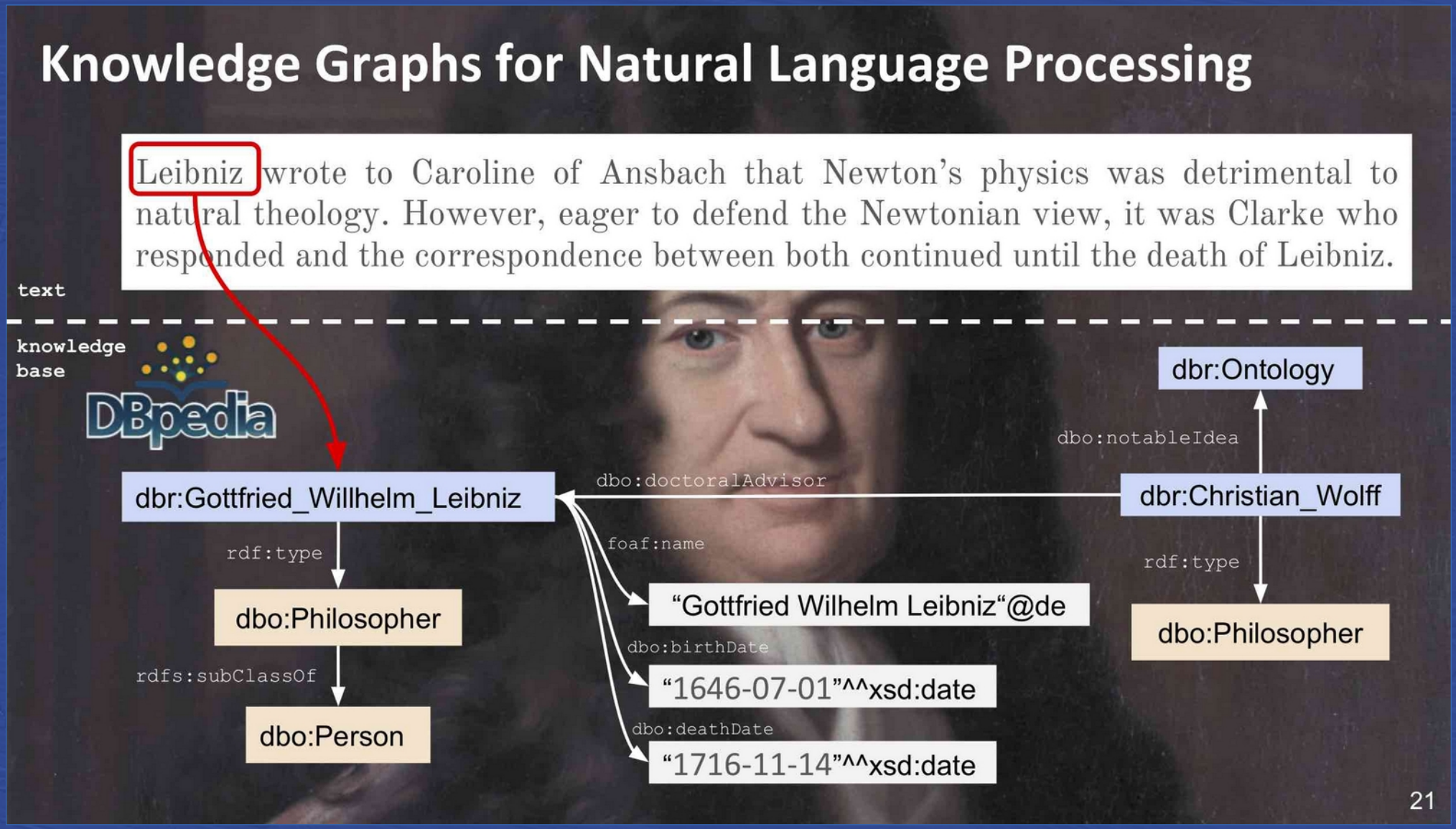
- « The Resource Description Framework (**RDF**) is a framework for **representing information in the Web**. »

- « A graph-based data model »



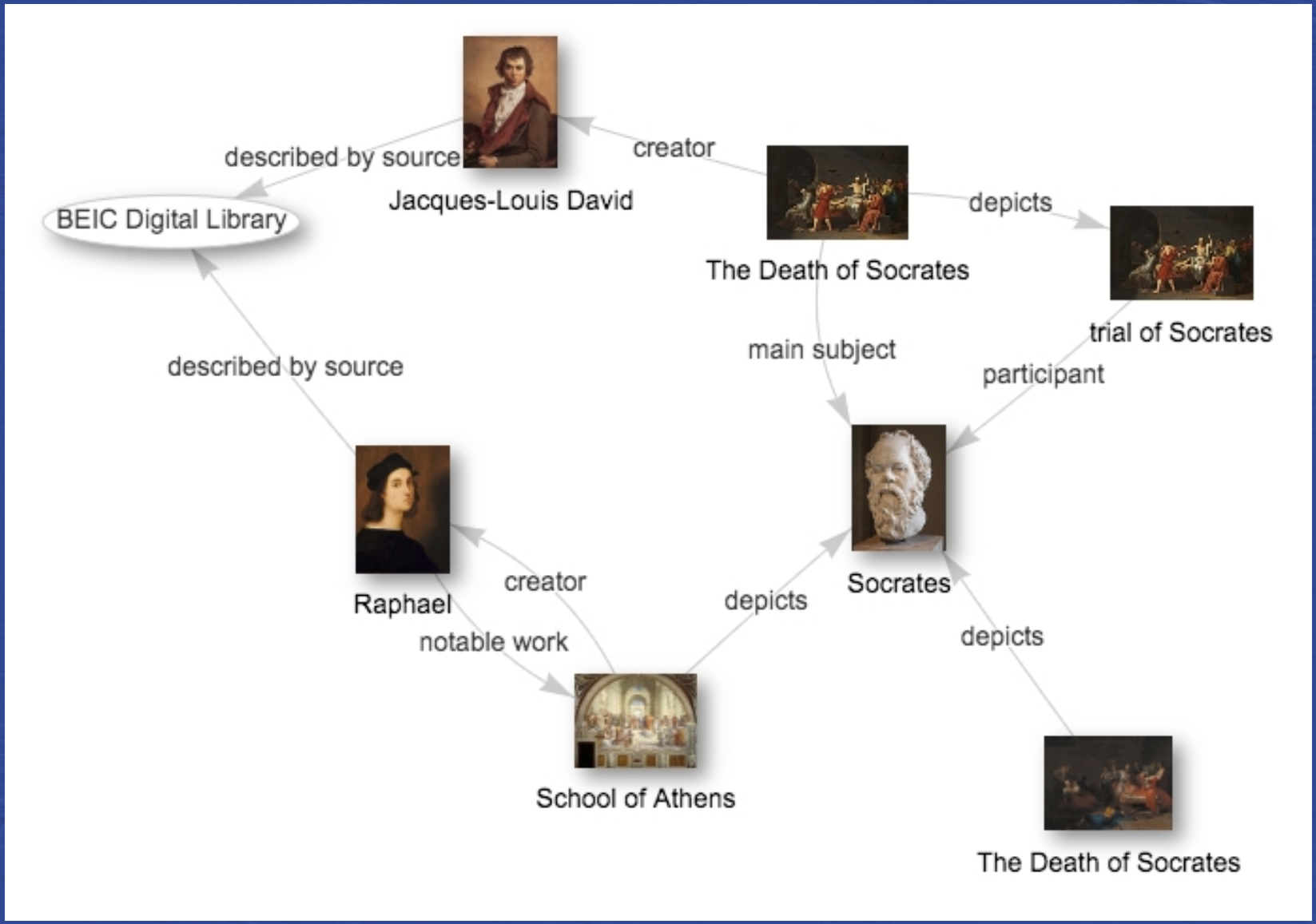
RDF triple

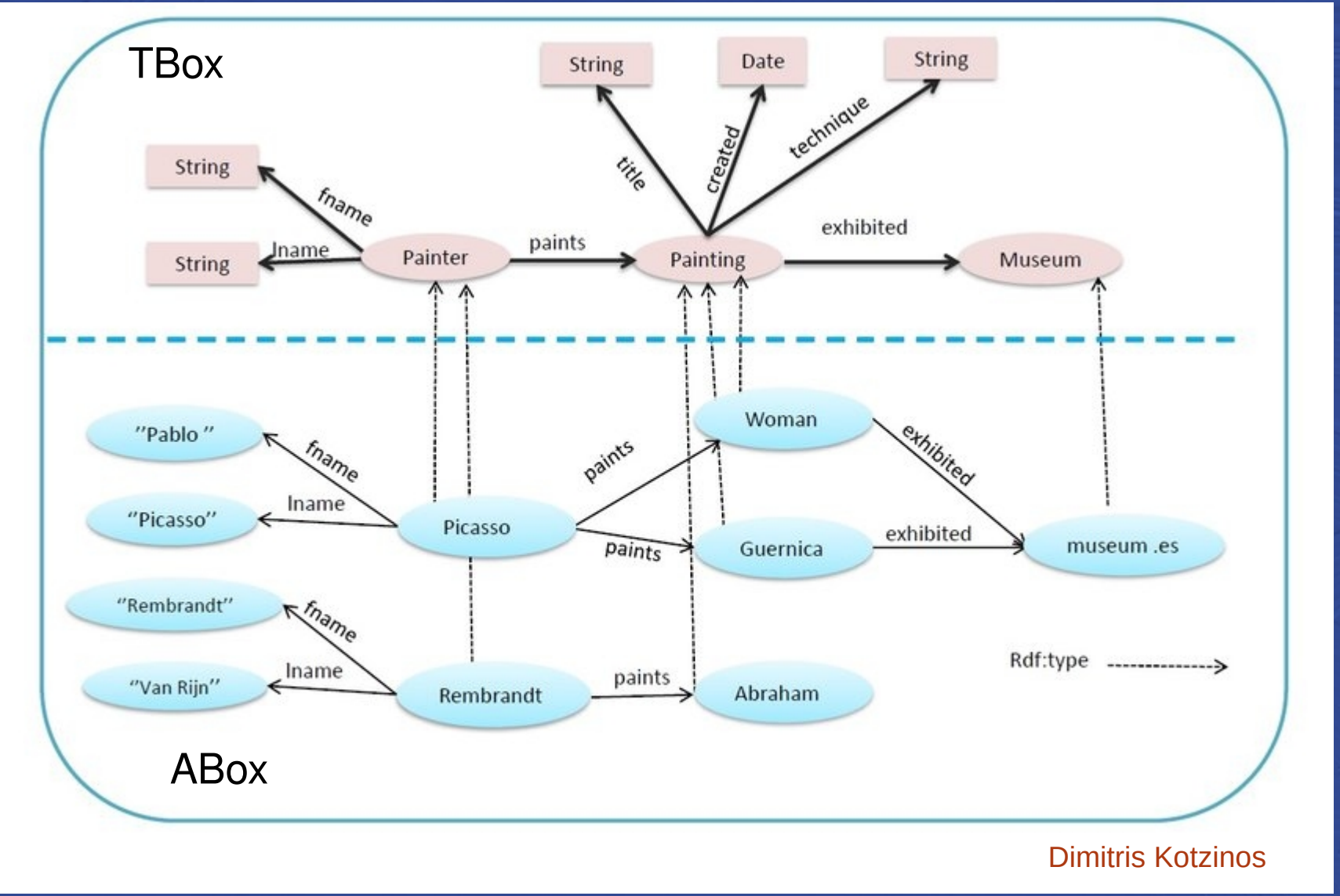
Where are the tables? In the graph !



Prof. Dr. Harald Sack, Antrittsvorlesung
Karlsruher Institut für Technologie (29.11.2017)

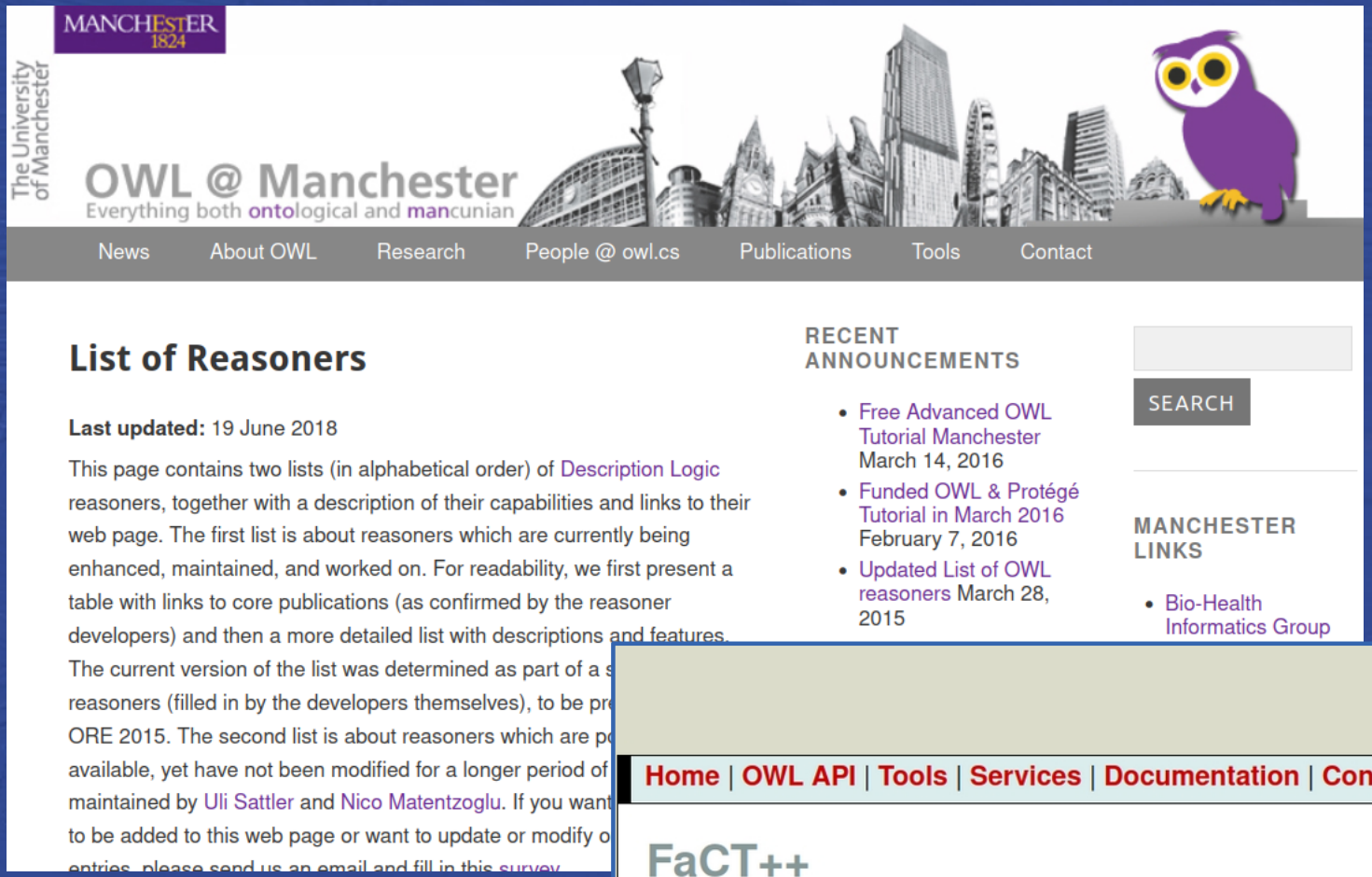
Wikidata 👈 An information graph (*knowledge graph*) representing the objects in the world and their relationships

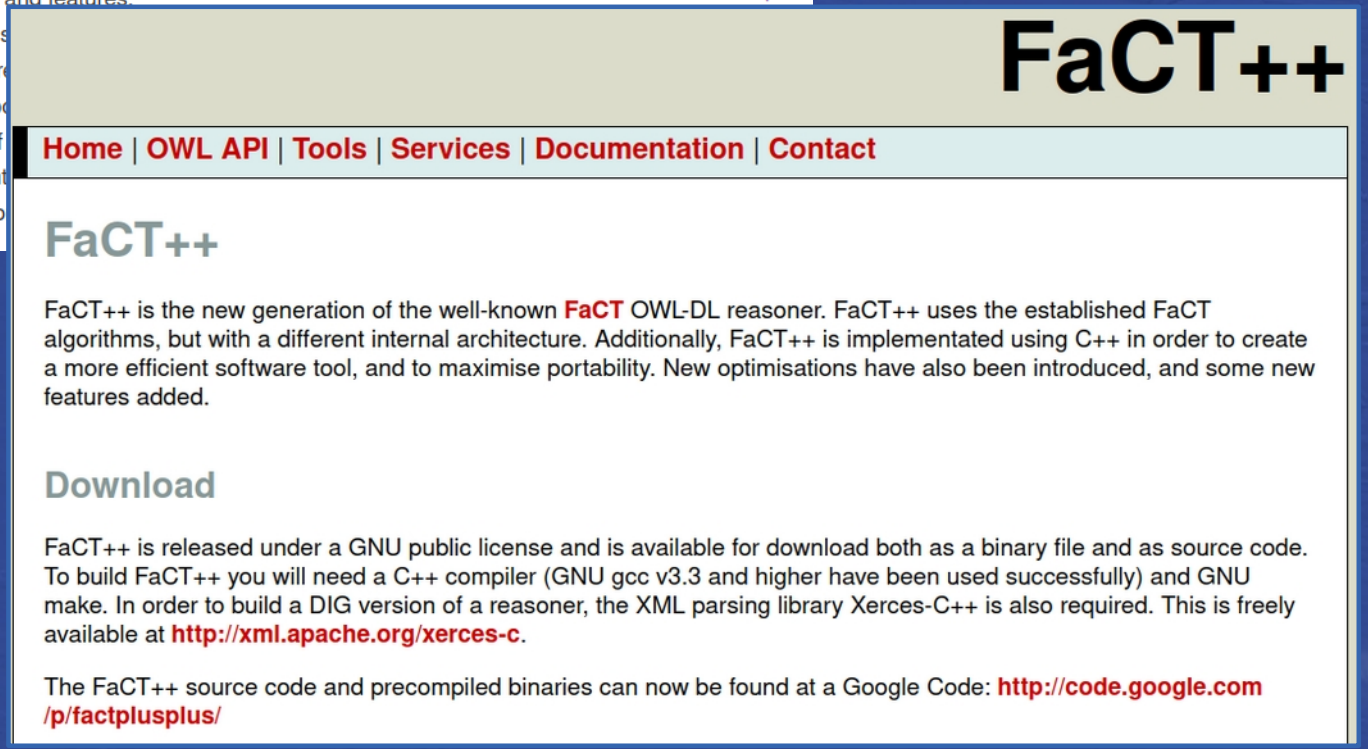




Dimitris Kotzinos

- ABox (Assertional Box) information graph, instance-level data
- TBox (Terminological Box): schema-level rules, RDFS / OWL-DL / SCHACL


MANCHESTER 1824
The University of Manchester
OWL @ Manchester
Everything both ontological and mancunian
News
About OWL
Research
People @ owl.cs
Publications
Tools
Contact
List of Reasoners
Last updated: 19 June 2018
This page contains two lists (in alphabetical order) of Description Logic reasoners, together with a description of their capabilities and links to their web page. The first list is about reasoners which are currently being enhanced, maintained, and worked on. For readability, we first present a table with links to core publications (as confirmed by the reasoner developers) and then a more detailed list with descriptions and features.
The current version of the list was determined as part of a s
reasoners (filled in by the developers themselves), to be pr
ORE 2015. The second list is about reasoners which are p
available, yet have not been modified for a longer period of
maintained by Uli Sattler and Nico Matentzoglu. If you want
to be added to this web page or want to update or modify o
entries, please send us an email and fill in this survey
RECENT ANNOUNCEMENTS
Free Advanced OWL Tutorial Manchester March 14, 2016
Funded OWL & Protégé Tutorial in March 2016 February 7, 2016
Updated List of OWL reasoners March 28, 2015
SEARCH
MANCHESTER LINKS
Bio-Health Informatics Group


FaCT++
Home | OWL API | Tools | Services | Documentation | Contact
FaCT++
FaCT++ is the new generation of the well-known FaCT OWL-DL reasoner. FaCT++ uses the established FaCT algorithms, but with a different internal architecture. Additionally, FaCT++ is implementated using C++ in order to create a more efficient software tool, and to maximise portability. New optimisations have also been introduced, and some new features added.
Download
FaCT++ is released under a GNU public license and is available for download both as a binary file and as source code. To build FaCT++ you will need a C++ compiler (GNU gcc v3.3 and higher have been used successfully) and GNU make. In order to build a DIG version of a reasoner, the XML parsing library Xerces-C++ is also required. This is freely available at http://xml.apache.org/xerces-c.
The FaCT++ source code and precompiled binaries can now be found at a Google Code: http://code.google.com /p/factplusplus/

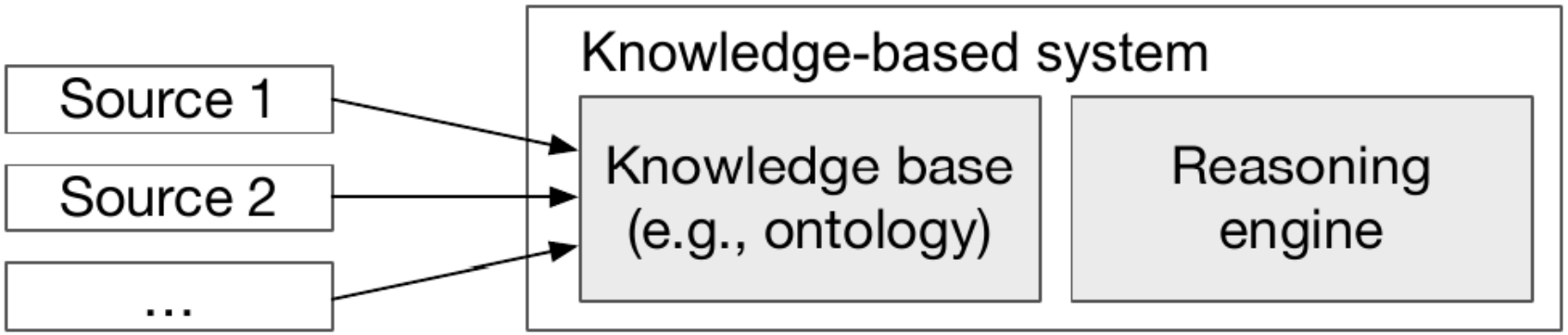


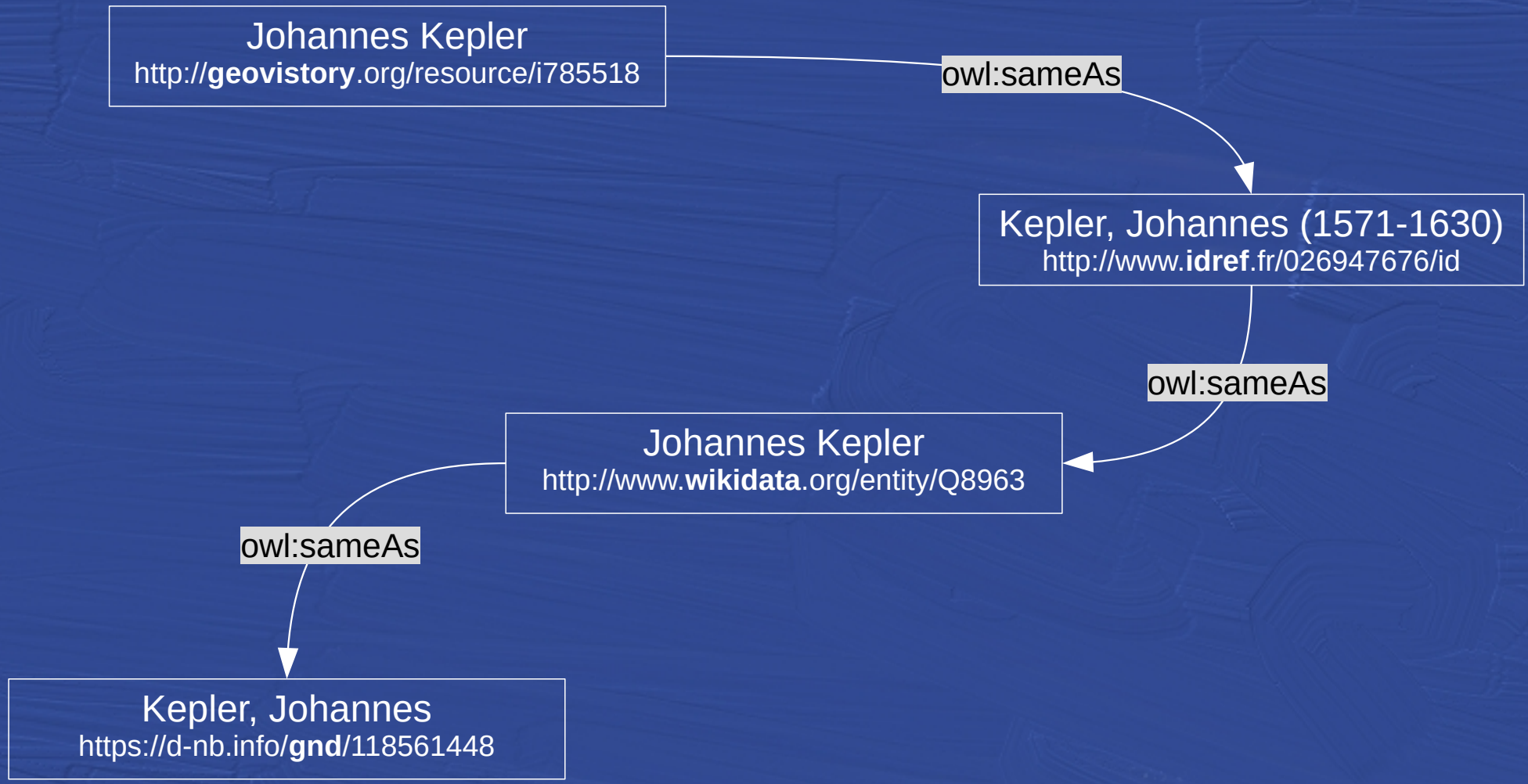
Figure 1: Architecture of a knowledge graph

Ehrlinger, Lisa and Wolfram Wöß, 'Towards a Definition of Knowledge Graphs',

Joint Proceedings of the Posters and Demos Track of the 12th International Conference on Semantic Systems ..., Leipzig, Germany, September 12-15, 2016, 2016 (https://ceur-ws.org/Vol-1695/paper4.pdf)

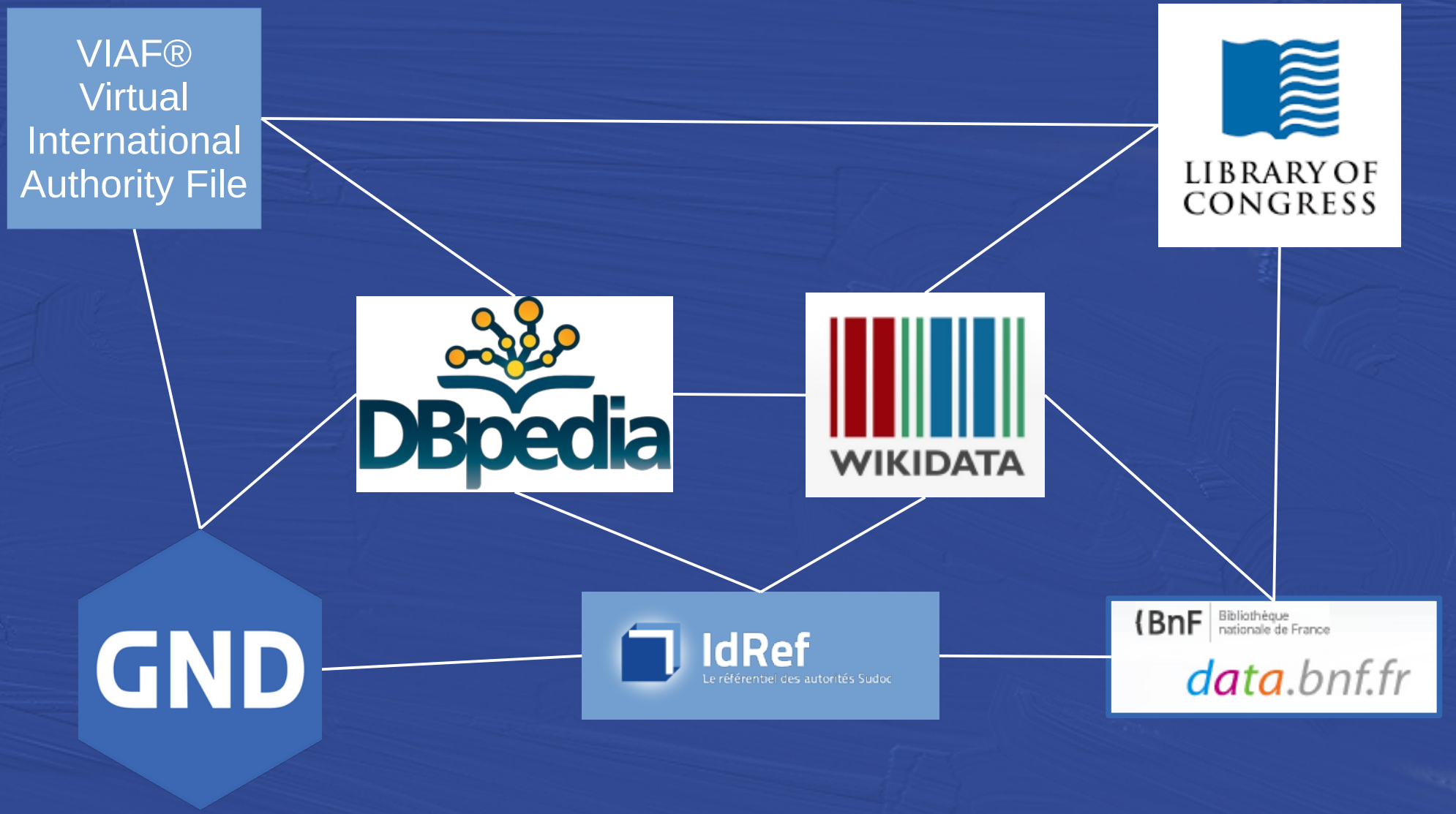
LOD as Links

« URIs to denote things, so that people can point at your stuff »

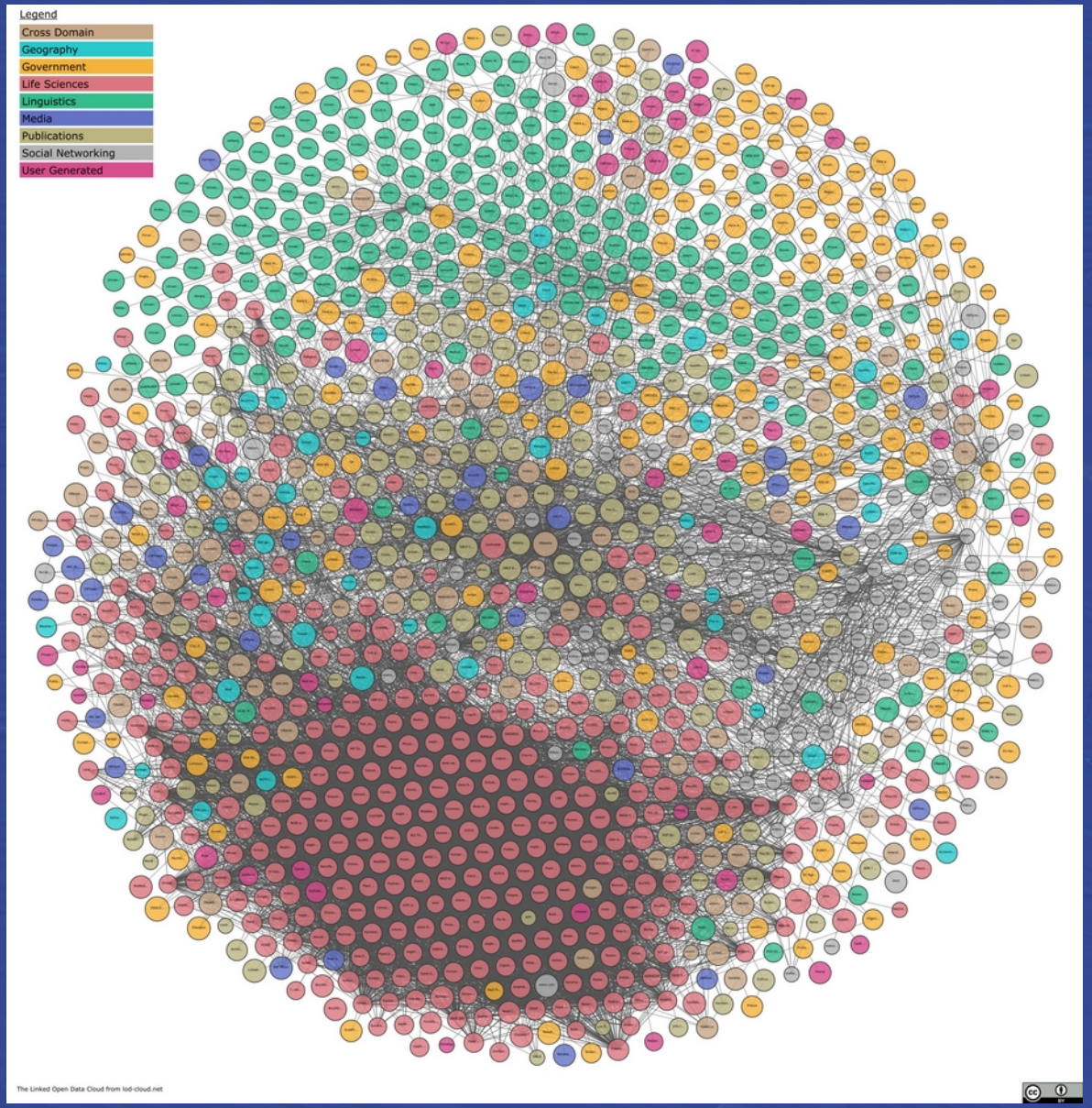


LOD as Links

« URIs to denote things, so that people can point at your stuff »

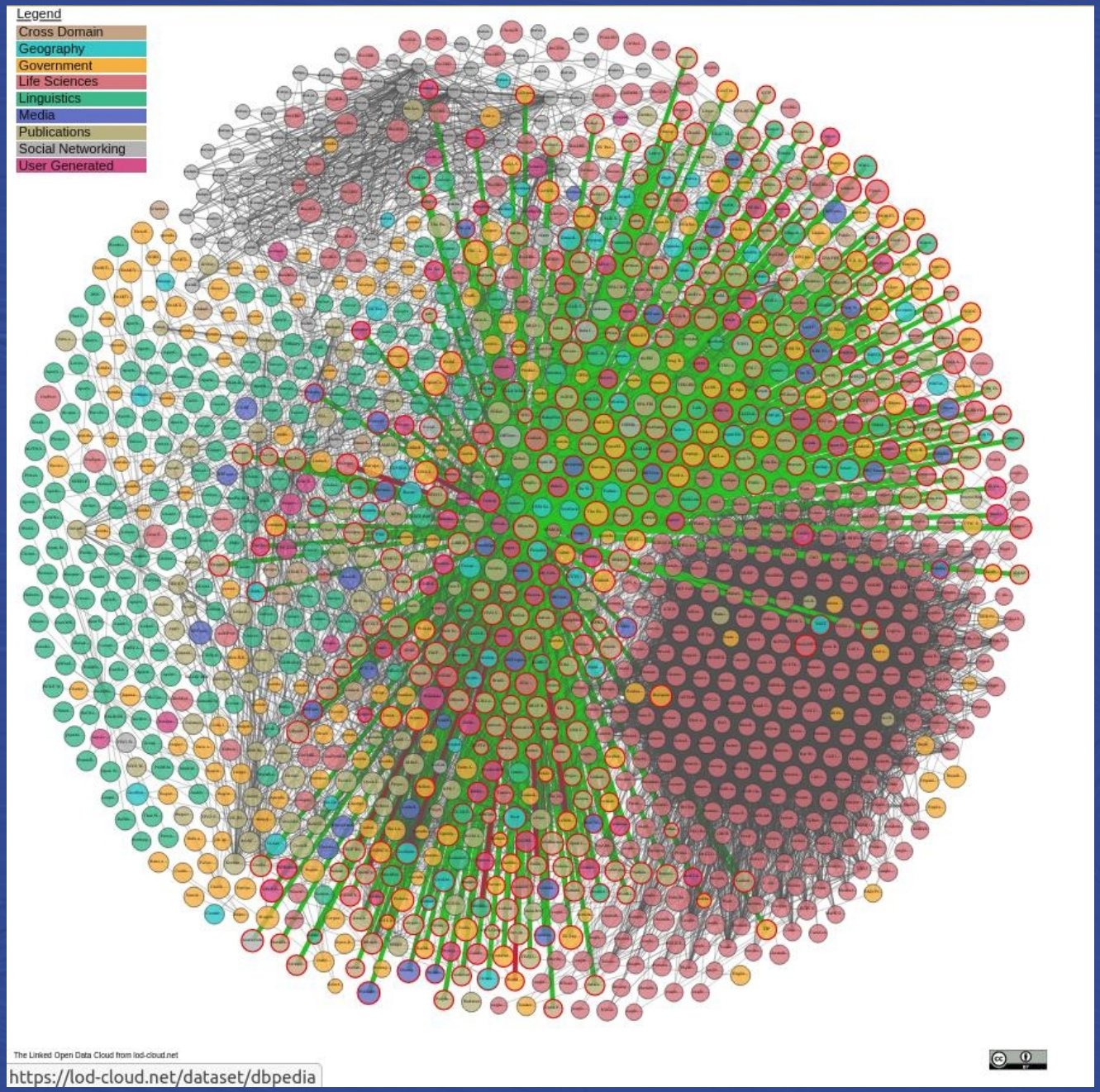


Semantic Web

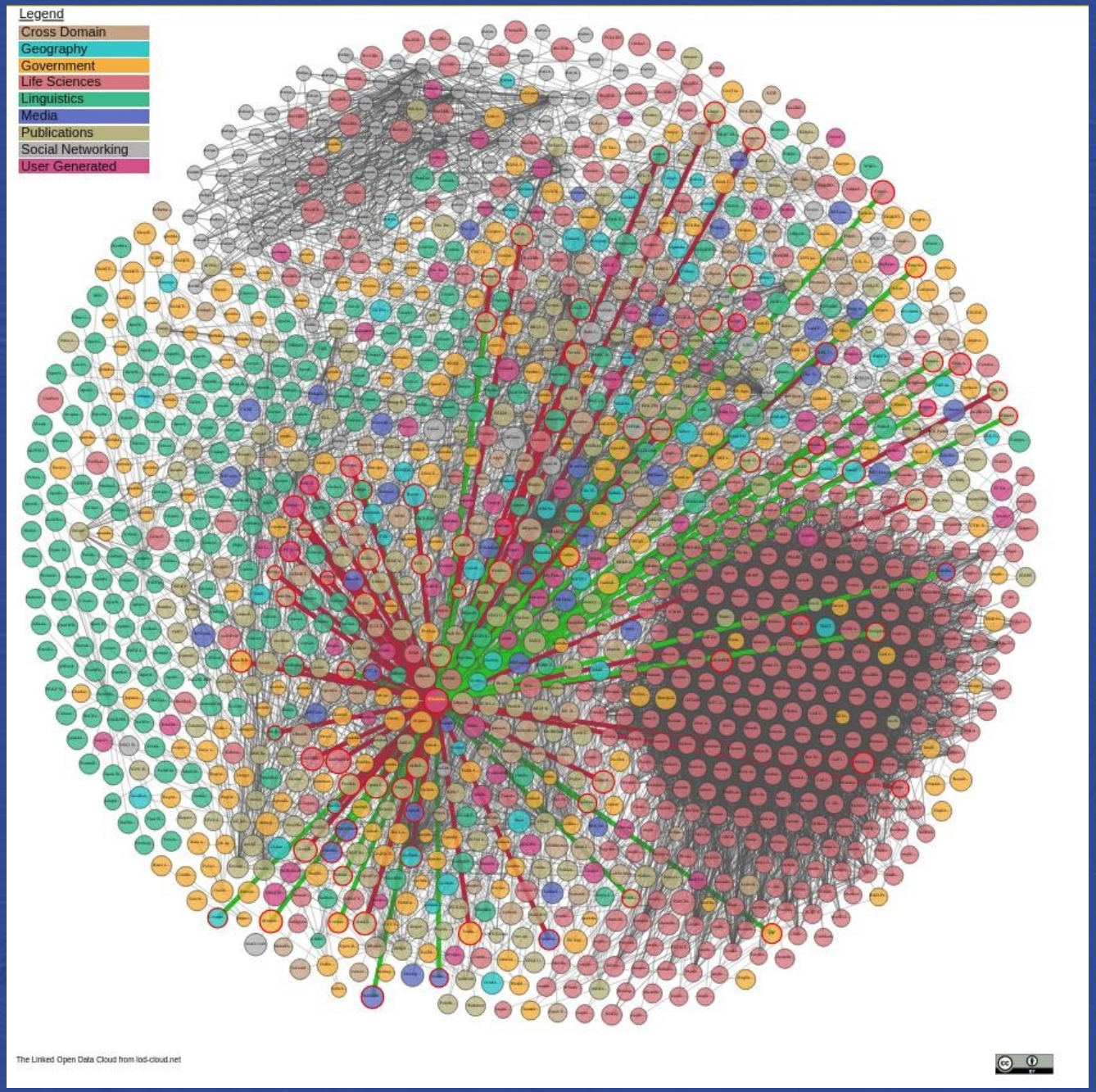


https://lod-cloud.net/

DBPedia

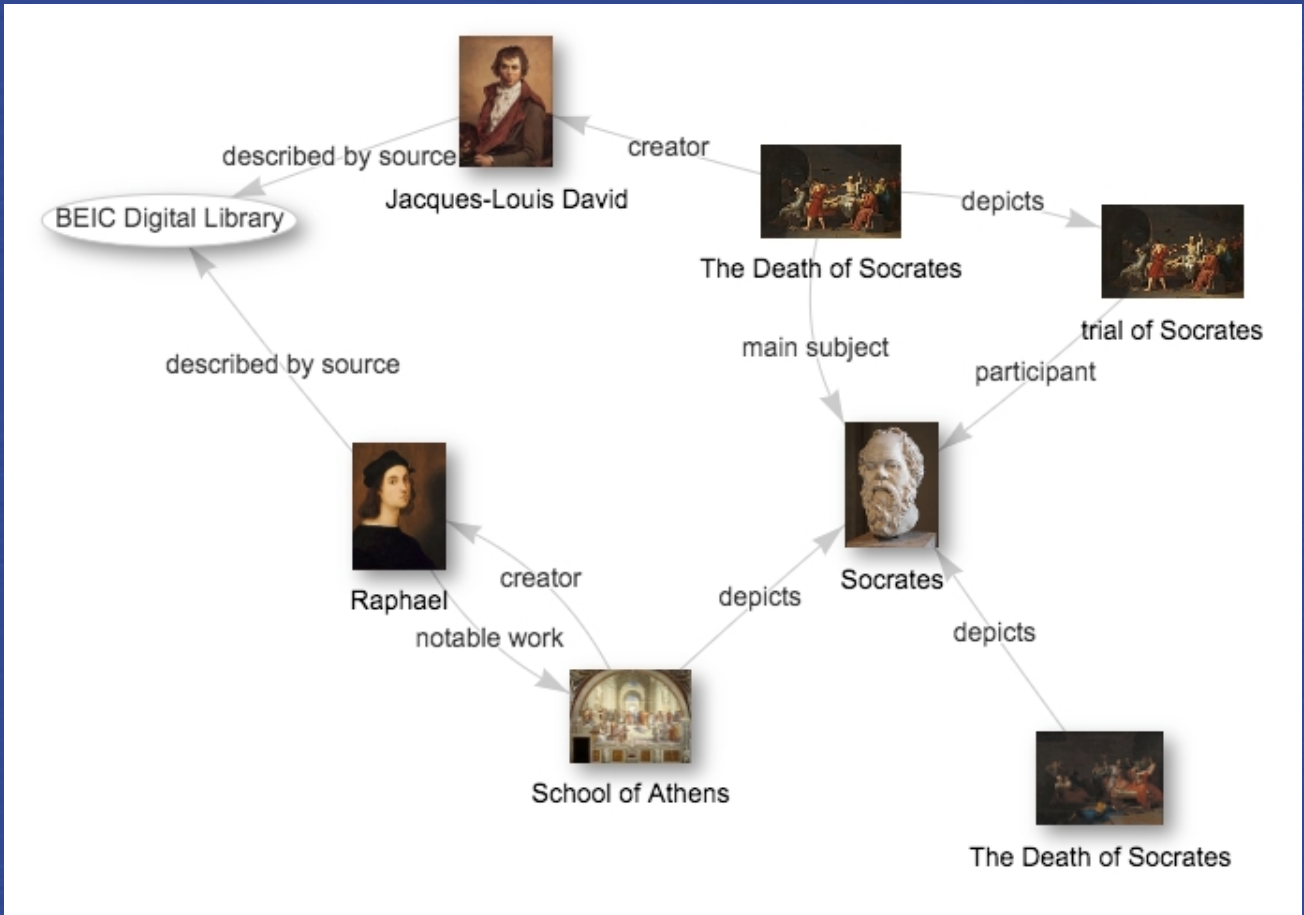

Legend
Cross Domain
Geography
Government
Life Sciences
Linguistics
Media
Publications
Social Networking
User Generated
The Linked Open Data Cloud from lod-cloud.net
https://lod-cloud.net/dataset/dbpedia

Wikidata


Legend
Cross Domain
Geography
Government
Life Sciences
Linguistics
Media
Publications
Social Networking
User Generated
The Linked Open Data Cloud from lod-cloud.net

LOD as a semantic graph connected through URIs

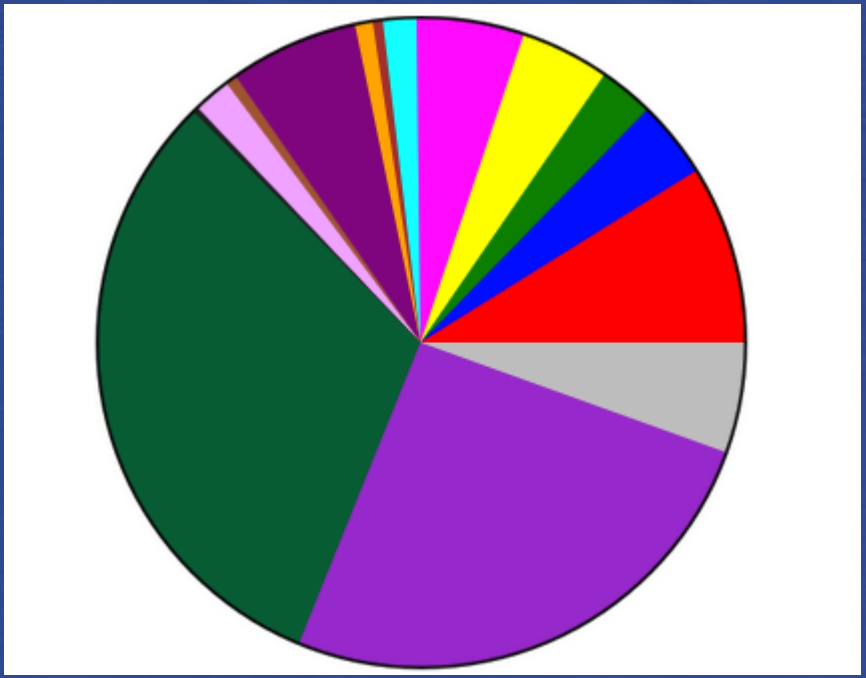
Wikidata as one of the central triplestores used to identify resources



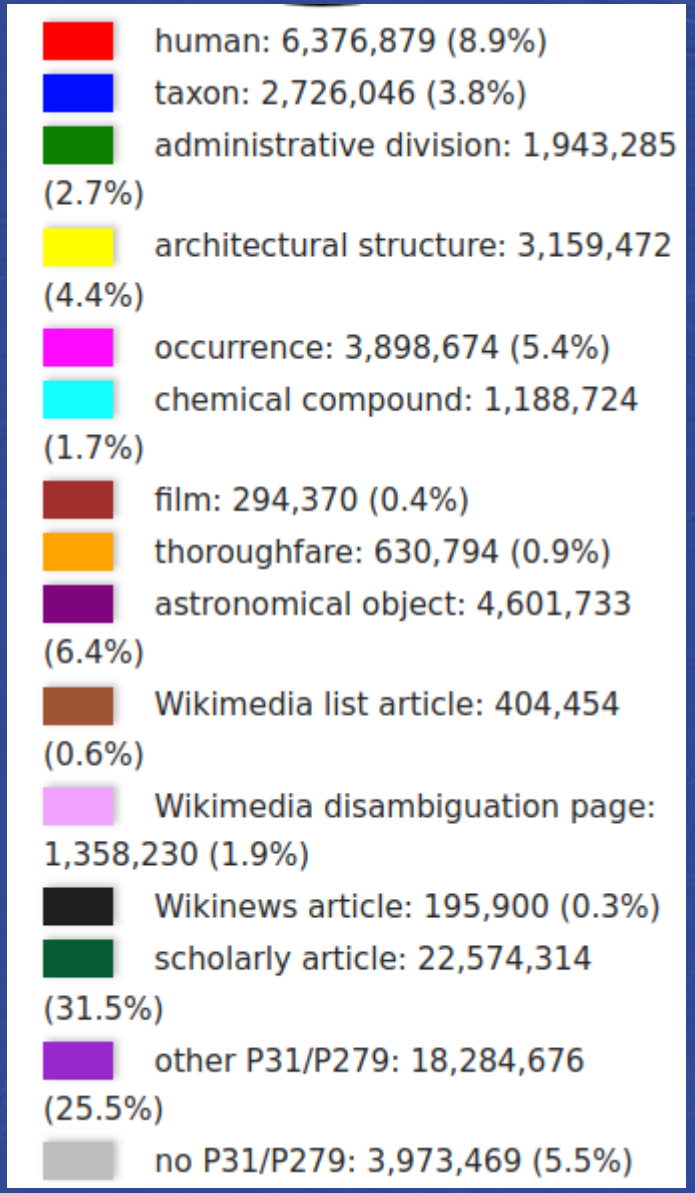
Wikidata

14 November 2023 – 107’588’216 items

1.5 billions statements

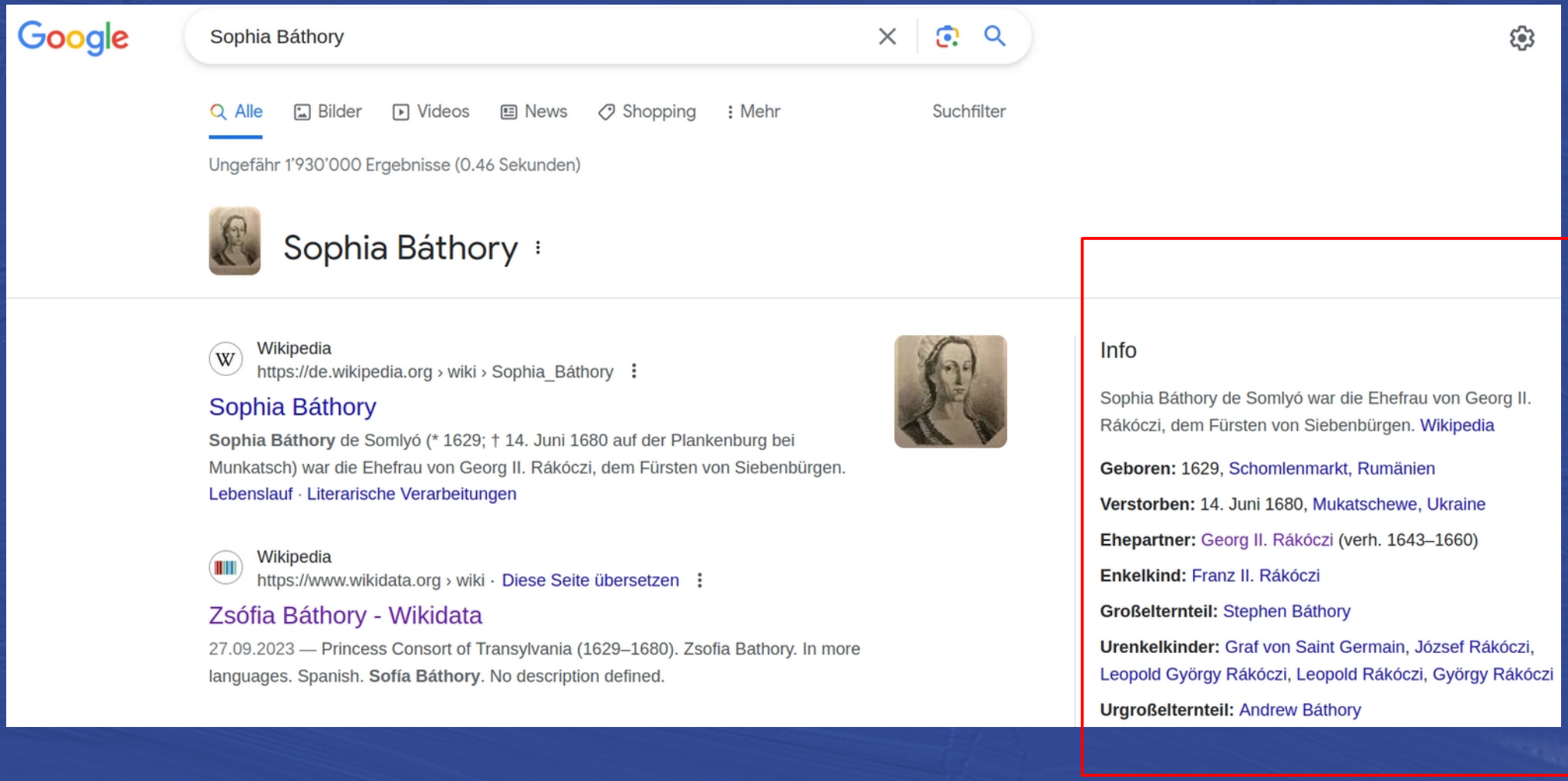


16 February 2020 : 71,611,020 items



Google Knowledge Graph

"By March 2023, it had grown to 800 billion facts on 8 billion entities" (Wikipedia).



Wikidata and Google Knowledge Graph

WIKIDATA

Item | Discussion

Zsófia Báthory (Q250942)

Princess Consort of Transylvania (1629–1680) edit

Zsofia Bathory

In more languages

Configure

Language	Label	Description	Also known as
English	Zsófia Báthory	Princess Consort of Transylvania (1629–1680)	Zsofia Bathory
German	Zsófia Báthory	Ehefrau von Georg II. Rákóczi, des Fürsten von Siebenbürgen (1629–1680)	
Alemannic	No label defined	No description defined	
French	Zsófia Báthory	(1629–1680)	

All entered languages

Main page
Community portal
Project chat
Create a new Item
Recent changes
Random Item
Query Service
Nearby
Help
Donate

Lexicographical data
Create a new Lexeme
Recent changes
Random Lexeme

Tools
What links here
Related changes
Special pages
Permanent link
Page information
Concept URI
Cite this page
Get shortened URL

Statements

instance of — human — 1 reference

image



Báthory Zsófia 1629.jpg
585 × 779; 155 KB

Google Knowledge Graph ID — /g/121258kx — edit
0 references
+ add reference
+ add value

Hungarian National Namespace person ID (new) — 662639 — edit
0 references
+ add reference
+ add value

WIKIDATA

Property | Discussion

Google Knowledge Graph ID (P2671)

identifier for Google Knowledge Graph API, starting with "/g/". For IDs starting with "/m/", use Freebase ID (P646)

In more languages

Configure

Language	Label	Description	Also known as
English	Google Knowledge Graph ID	identifier for Google Knowledge Graph API, starting with "/g/". For IDs starting with "/m/", use Freebase ID (P646)	

Main page
Community portal
Project chat
Create a new Item
Recent changes
Random Item
Query Service
Nearby

From a resource-centered model...



...to a statement-centered model...



WIKIDATA

BNF Authorities

statement

"1571-12-21"

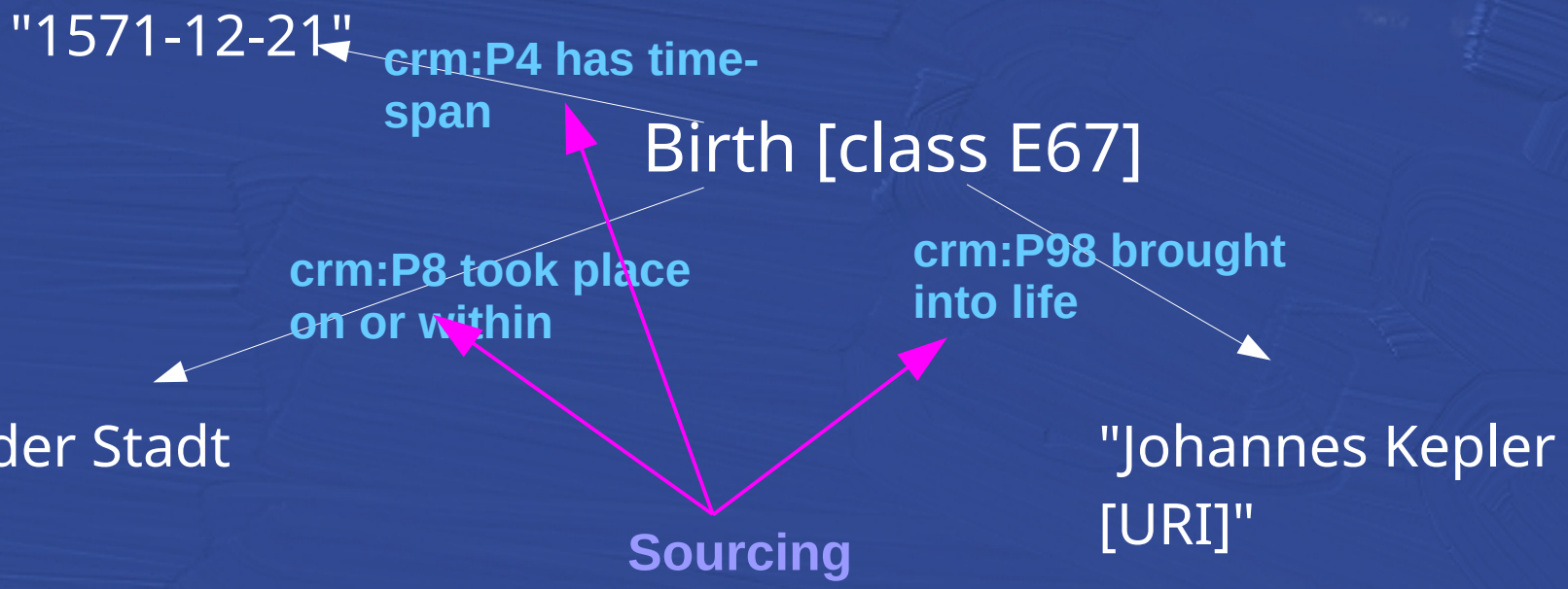
wdt:P569 (date of birth)

Kepler, Johannes (Q8963)

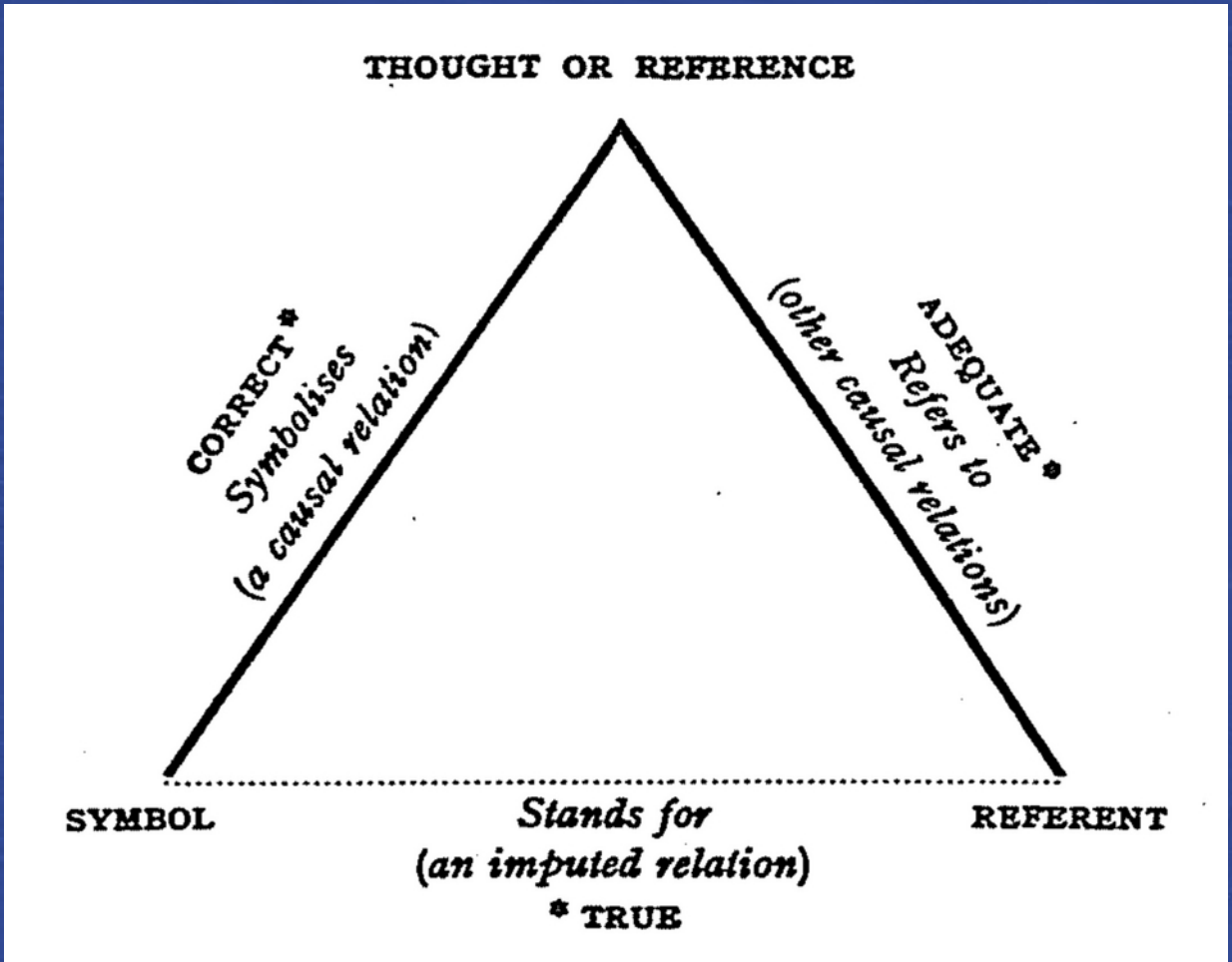
wdt:P19 (place of birth)

Weil der Stadt (Q505147)

… or prefer a more robust event-centered model !



Ontology - ontologies



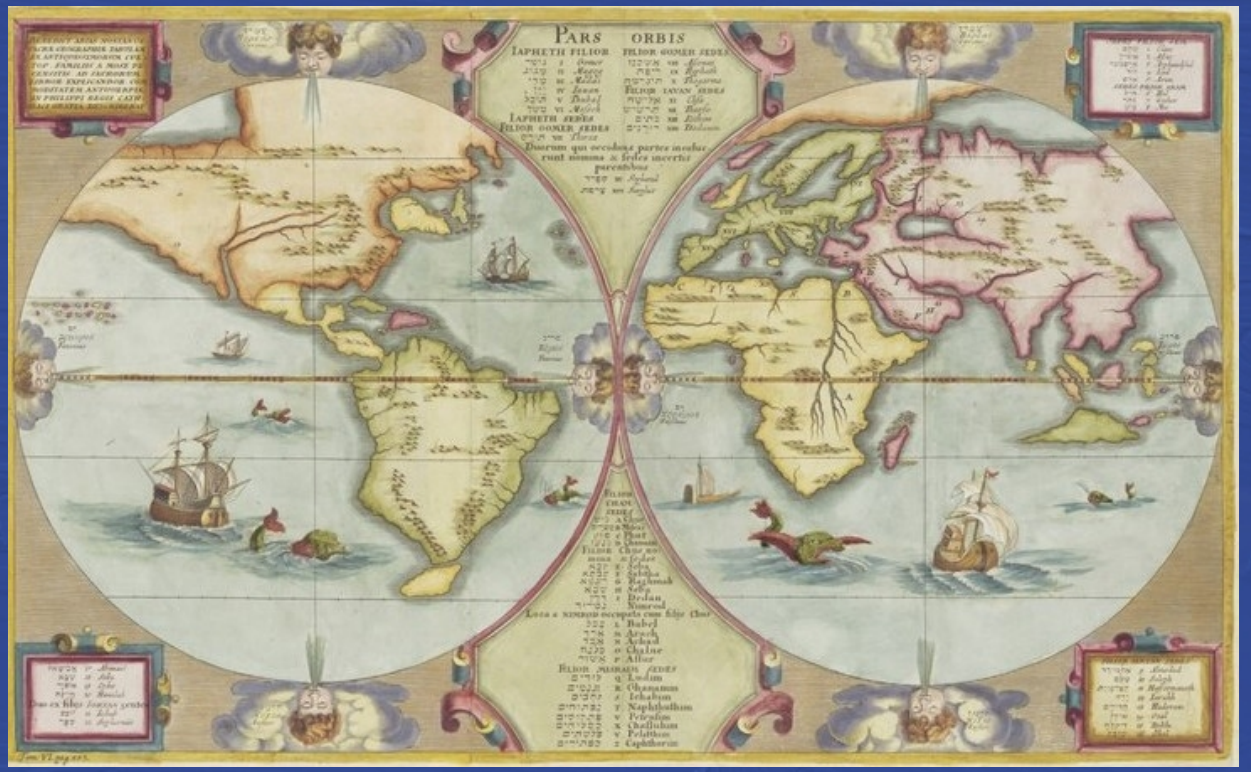
Triangle of reference

“An ontology is
a formal explicit specification
of a shared conceptualization
of a domain of interest”

- « Formality – ... a knowledge representation language that is based on the grounds of **formal semantics**. »
- « Consensus – … an agreement on a domain conceptualization among people in a community. »
- « Conceptuality – … in terms of conceptual symbols that can be intuitively grasped by humans, as they correspond to the elements in their **mental models**. »
- « Domain Specificity – ... limited to knowledge about a particular **domain of interest**. »

[Domingue et al. 2011, p. 510-511]

What is represented in the knowledge bases produced by HSS research?


PARS ORBIS

CC 0

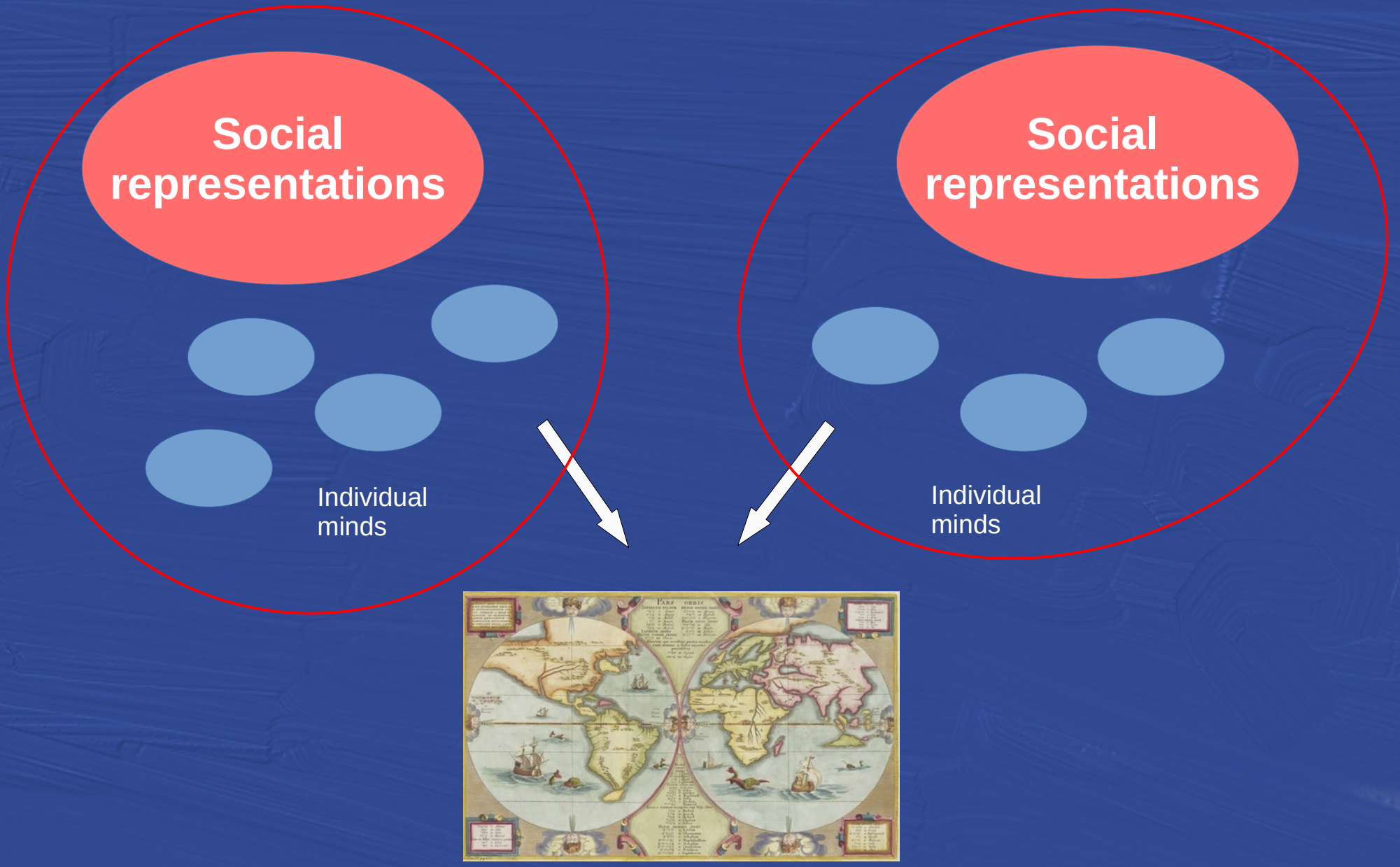




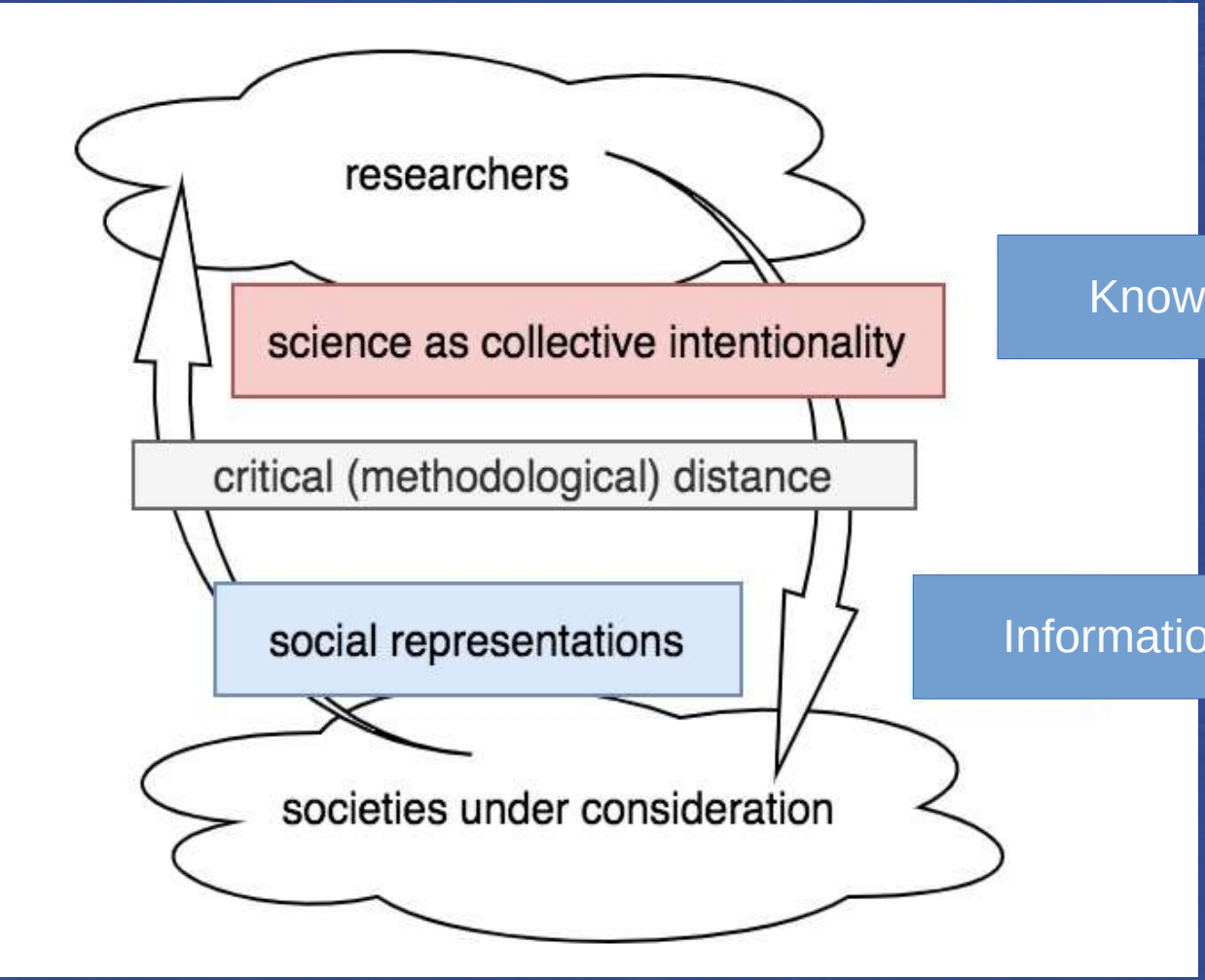

ATTEND
OBSERVE
LET GO
WELCOME
SLOW DOWN
NOTICE
PAUSE
RELEASE
LISTEN
TRUST
BE WITH
EMBRACE
VALUE
ADMIT
SENSE
ACCEPT
INVITE
OPEN
ALLOW

CC 0

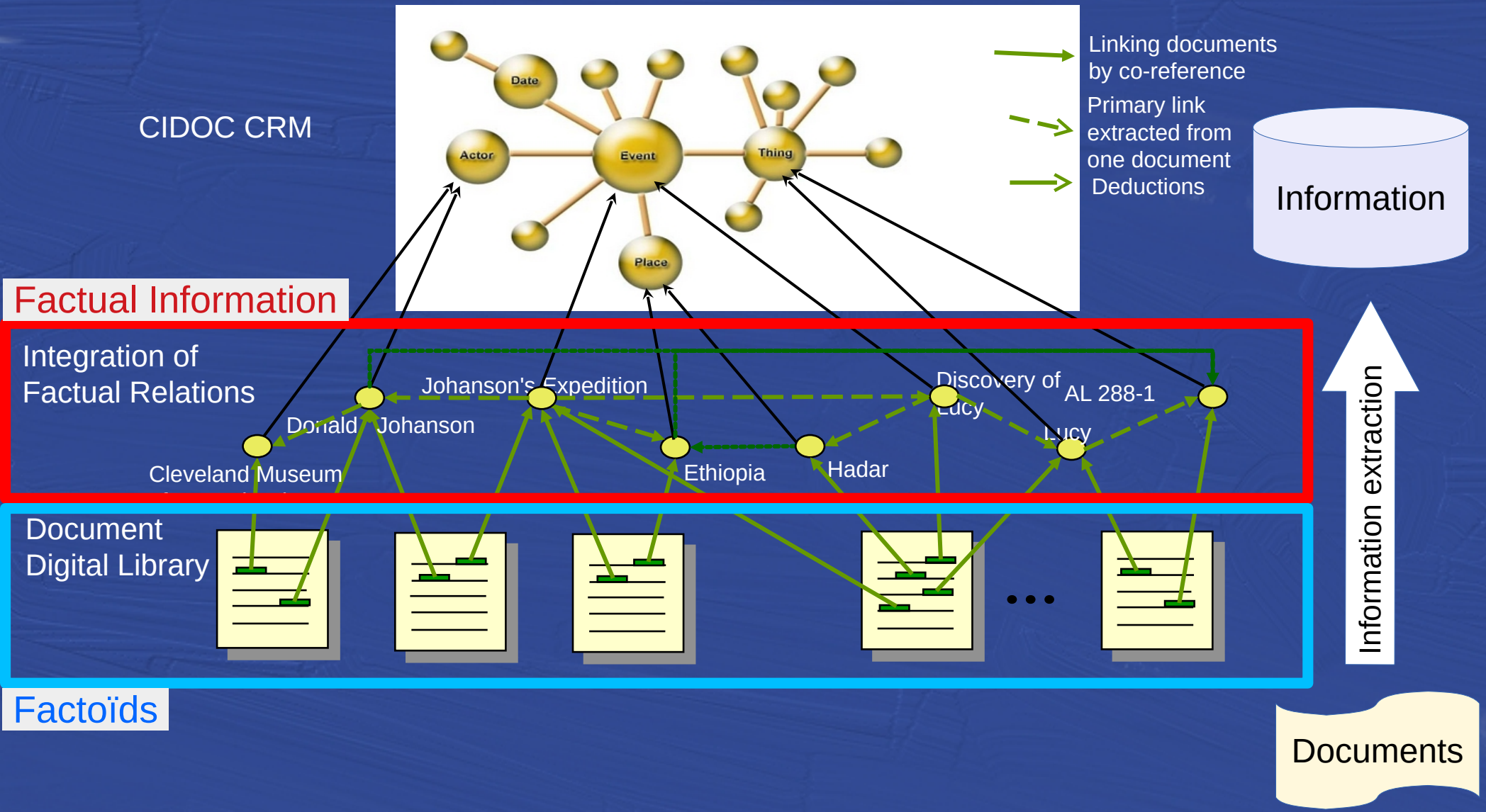
The observed phenomenon: different mental and social representations of the same material phenomena

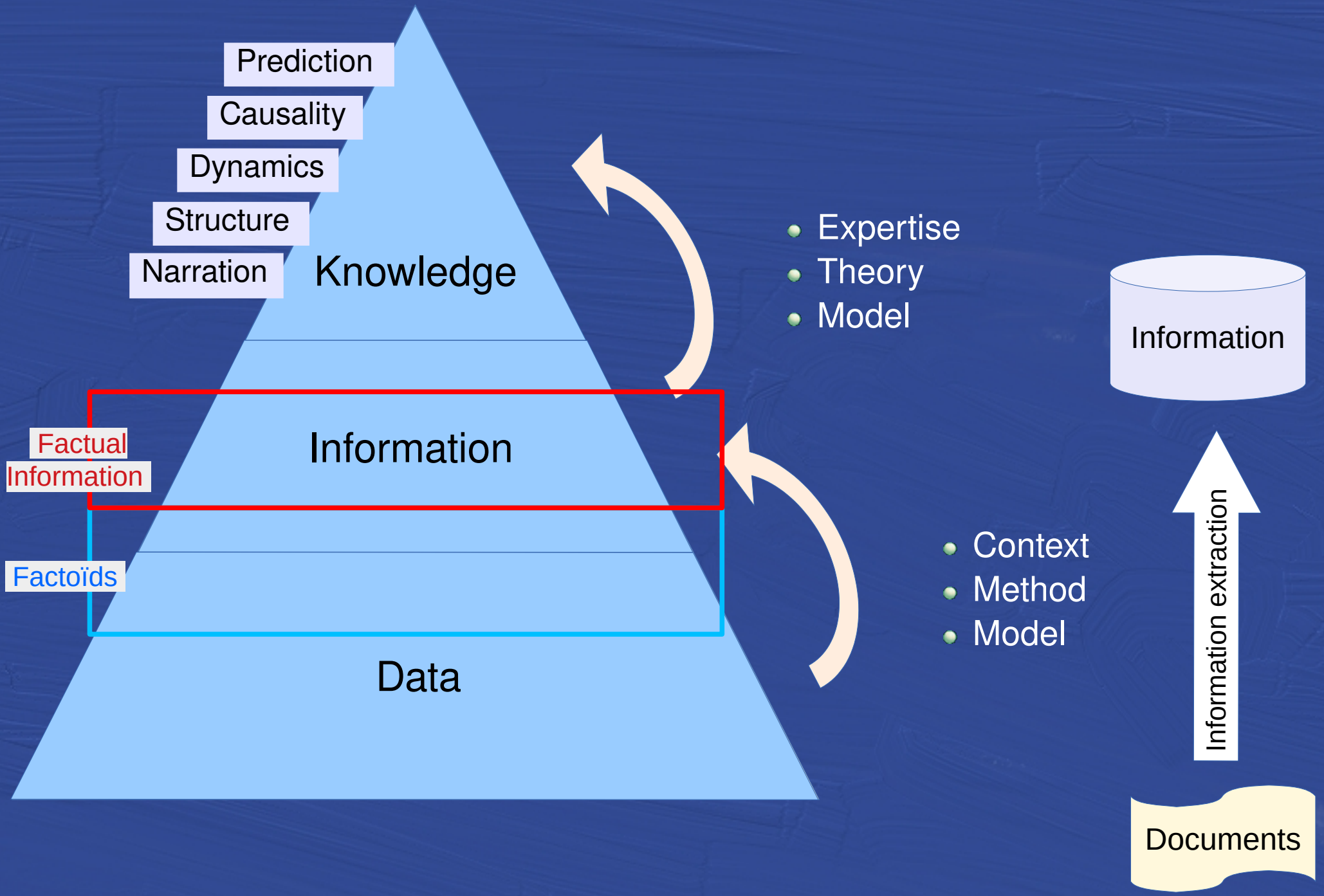


Information as representations of the world
Knowledge as representation of the world



Integration of information extracted from documents using the CIDOC CRM




Prediction
Causality
Dynamics
Structure
Narration
Knowledge
Information
Factual Information
Factoïds
Data
Expertise
Theory
Model
Context
Method
Model
Information
Information extraction
Documents

What Semantics for the information in the Semantic Web ?

Triangle of reference !!!

OntoClean

N. Guarino/C. A. Welty, « An Overview of OntoClean », in Steffen Staab, ed., Handbook on ontologies,
2nd ed. Berlin: Springer, 2009.

OntoClean (Wikipedia)

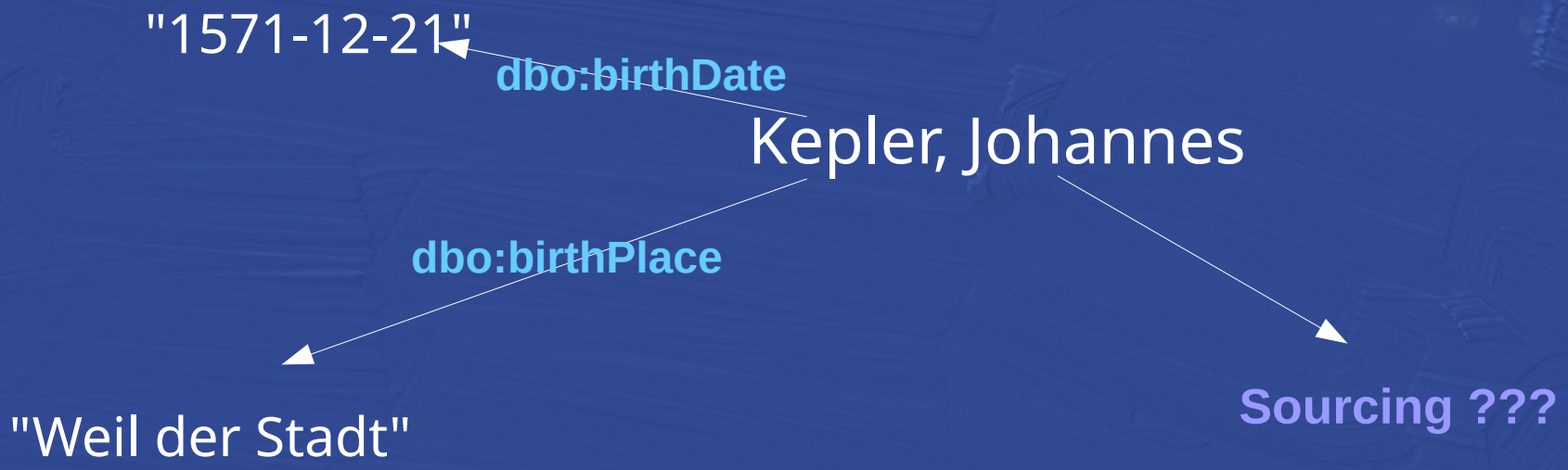
- Rigidity (Essence -> substance and accidents)
- Identity (Properties allowing to distinguish between individuals, e.g. events)
- Unity (Parthood : water, oceans)

A ‘student’ : not a class but a time-indexed property of a person.

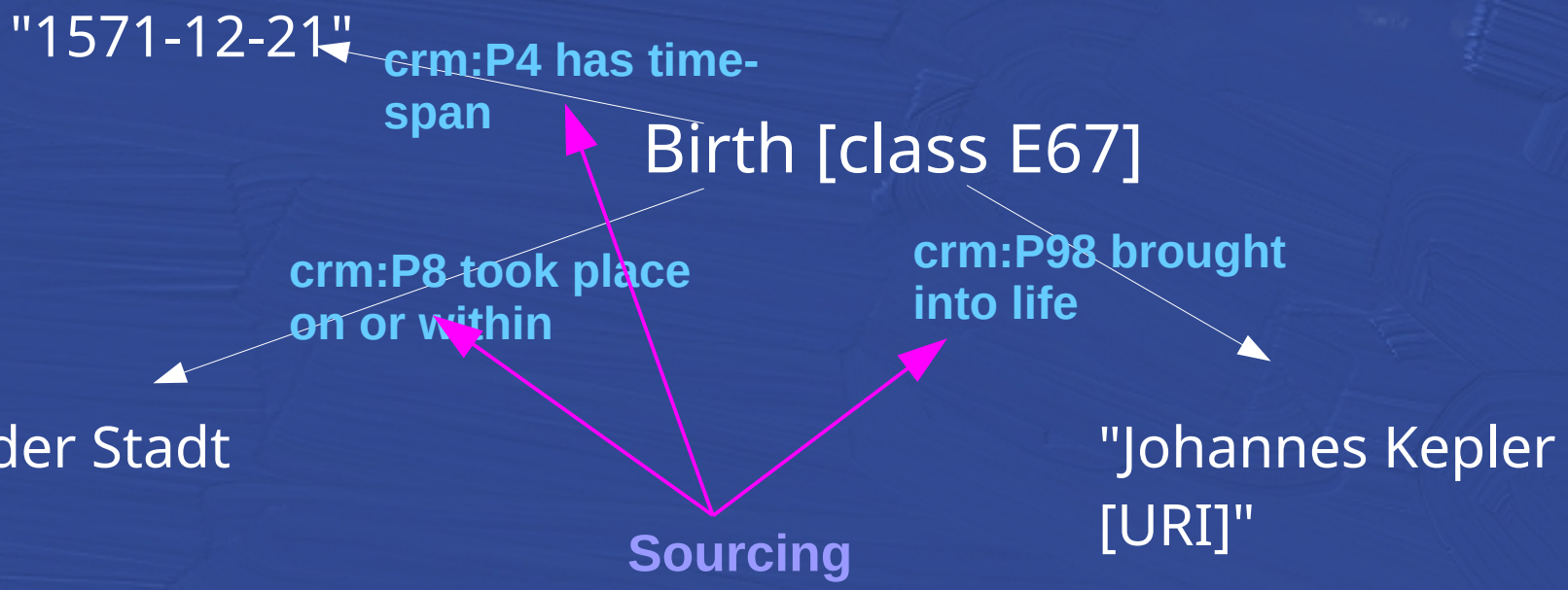
Modelling Best Practices

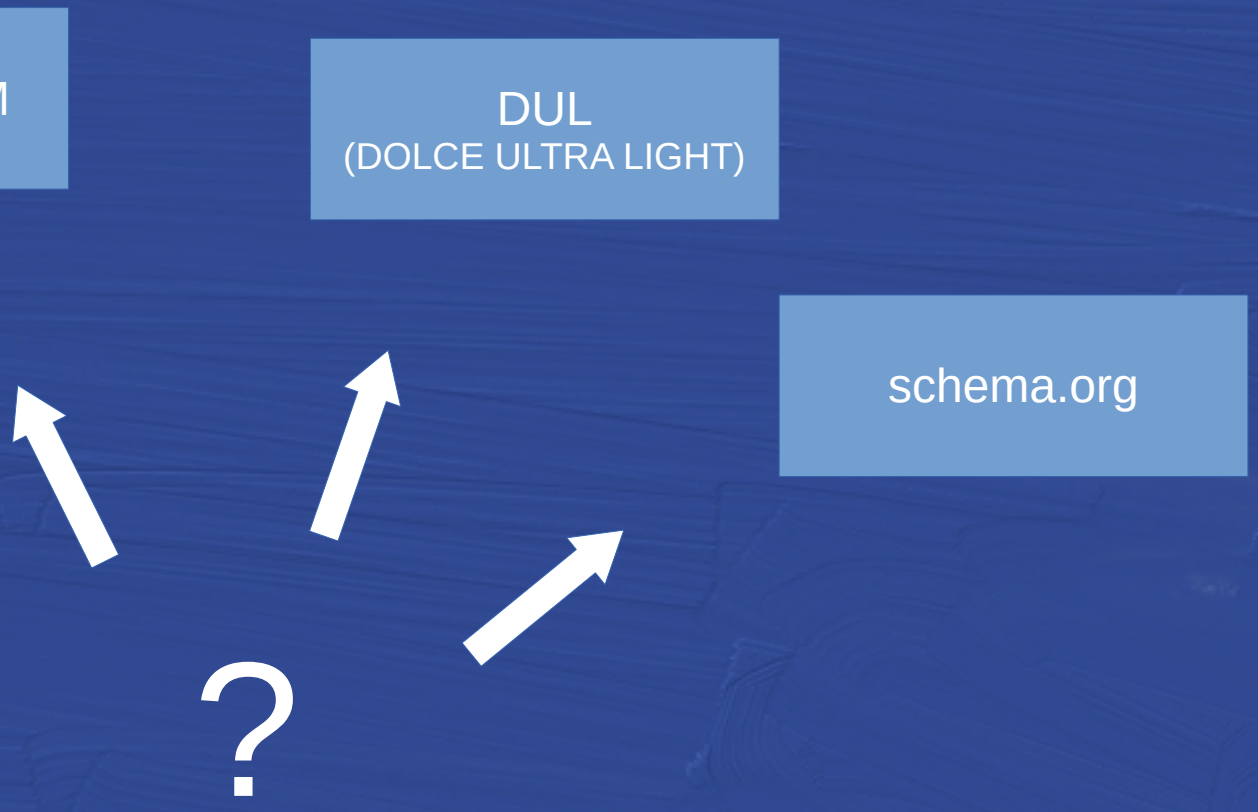
- Instances (research related, detailed) and Classes (few and cleanly defined)
- Controlled Vocabularies (researchers) and Ontologies (semantic engineers)
- Property Inheritance, Quantifiers

Adopt a resource-centered model...



… or prefer a more robust event-centered model !





Research agenda

Research specific data model

Research data

Foundational ontologies
& modelling best practices

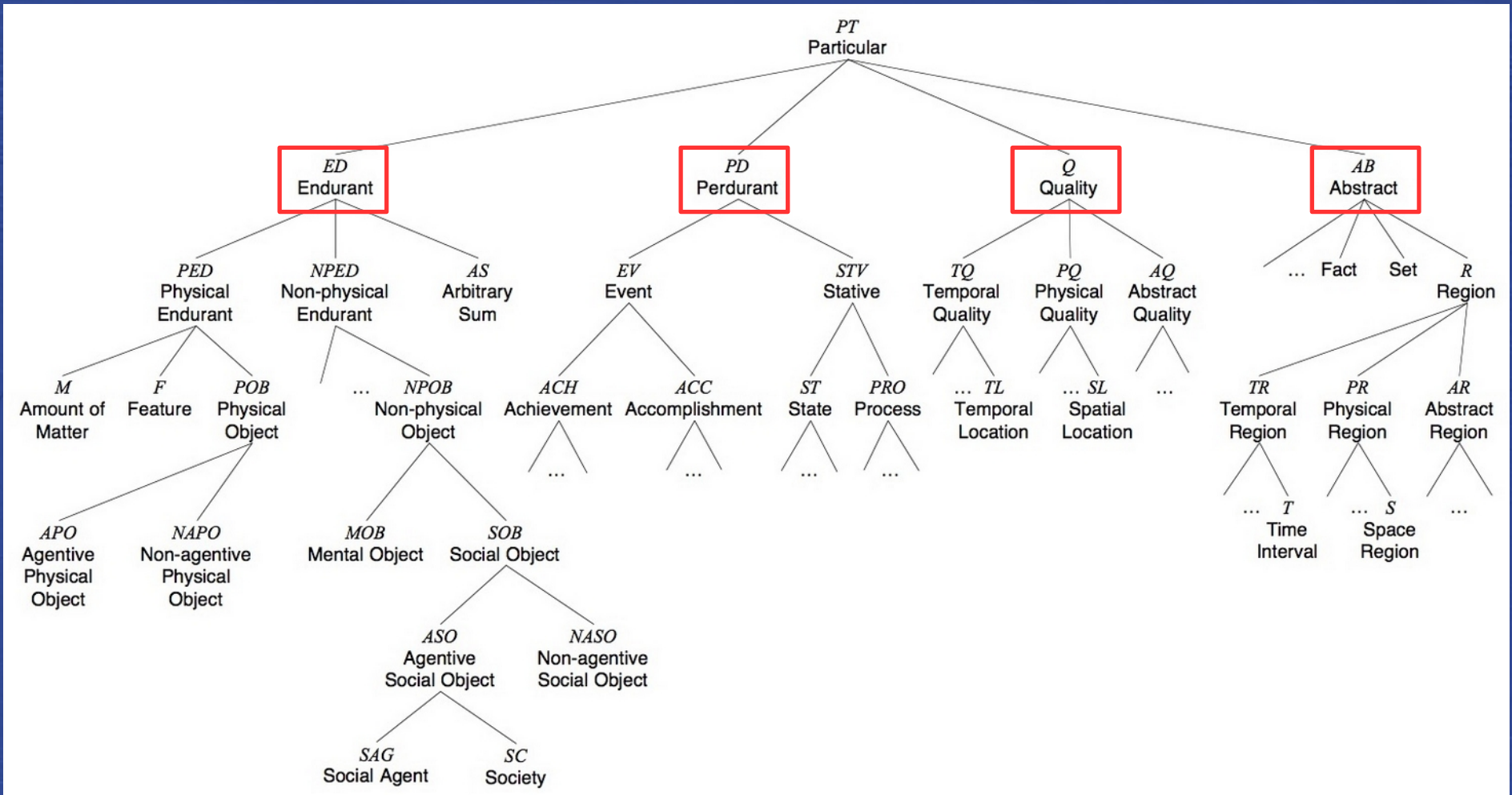
DOLCE + Descriptions and Situations
& object-oriented modelling principles

Research agenda

Research specific data model

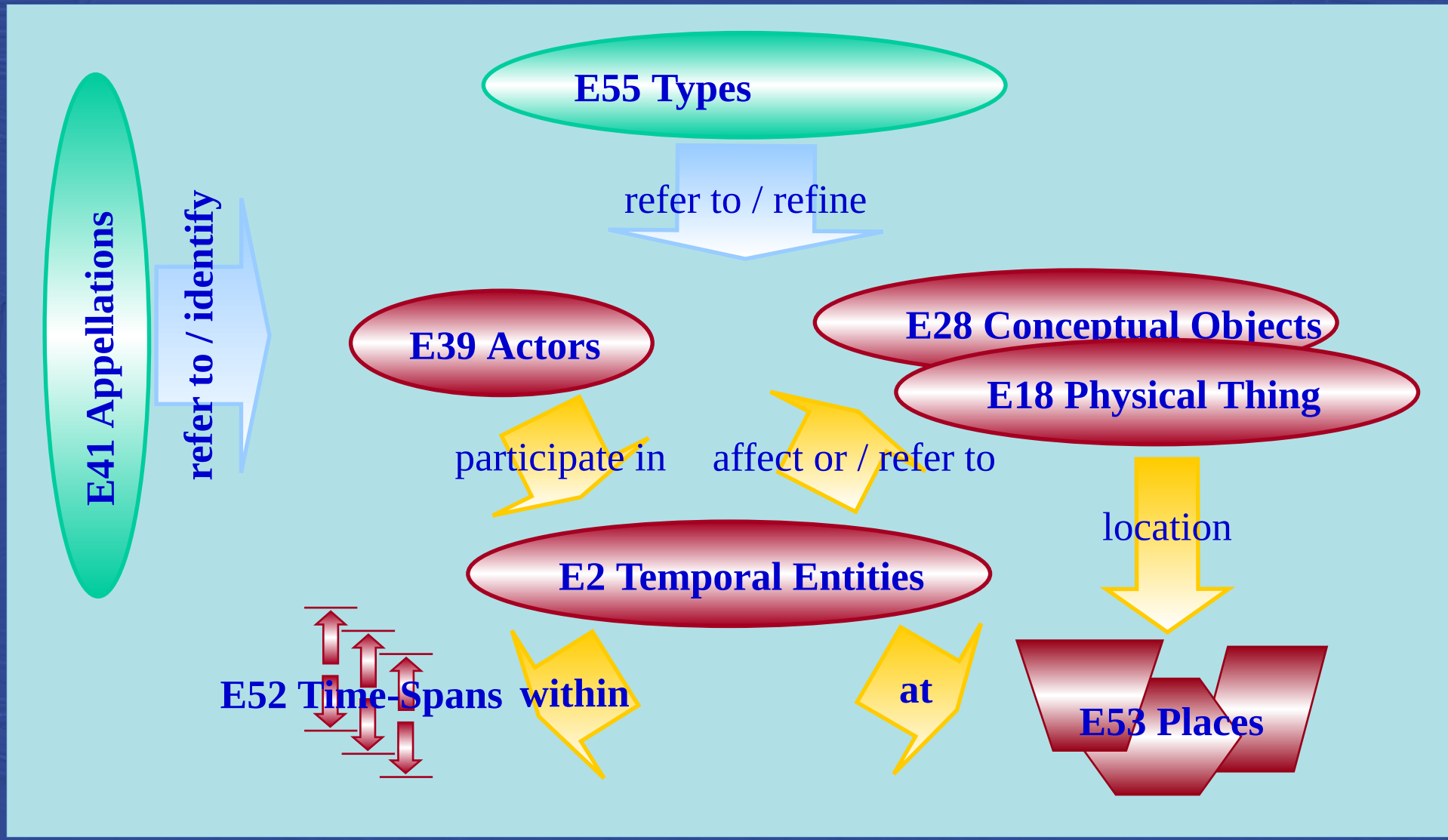
Research data

Foundational ontologies
were developed to support the verification and improvement
of the **conceptualization** of a **domain of discourse.**



Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) – a foundational ontology designed in 2002 in the context of the WonderWeb EU project, developed by Nicola Guarino and his associates at the Laboratory for Applied Ontology (LOA) – WonderWeb Deliverable D18, p.14

The CIDOC CRM (ISO21127:2006)
A semantic framework that provides *interoperability*
between different sources of **cultural heritage information**



Stephen Stead (2008)

Foundational ontologies
& modelling best practices

DOLCE + Descriptions and Situations
& object-oriented modelling principles





Generic, domain related core ontology

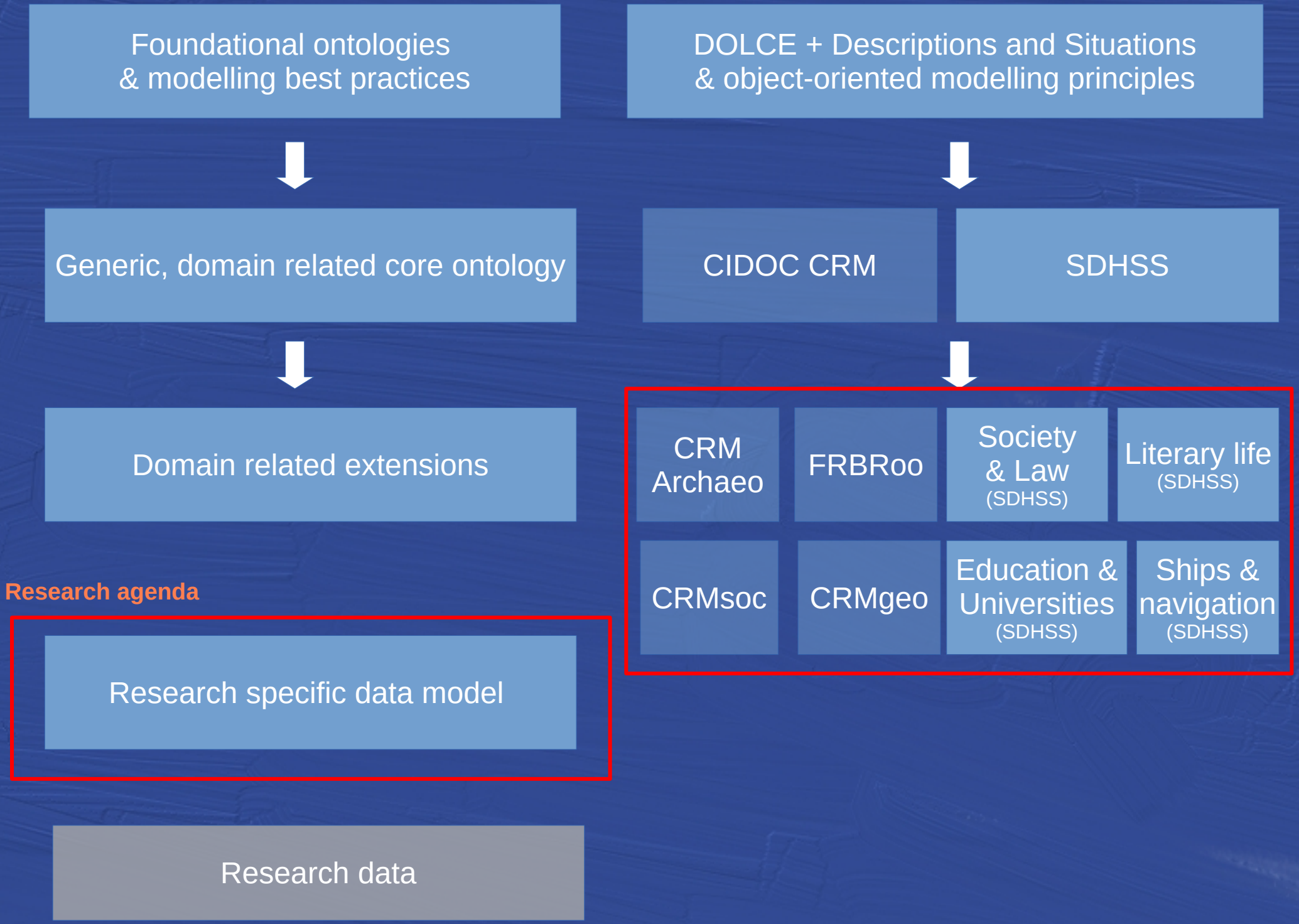
CIDOC CRM

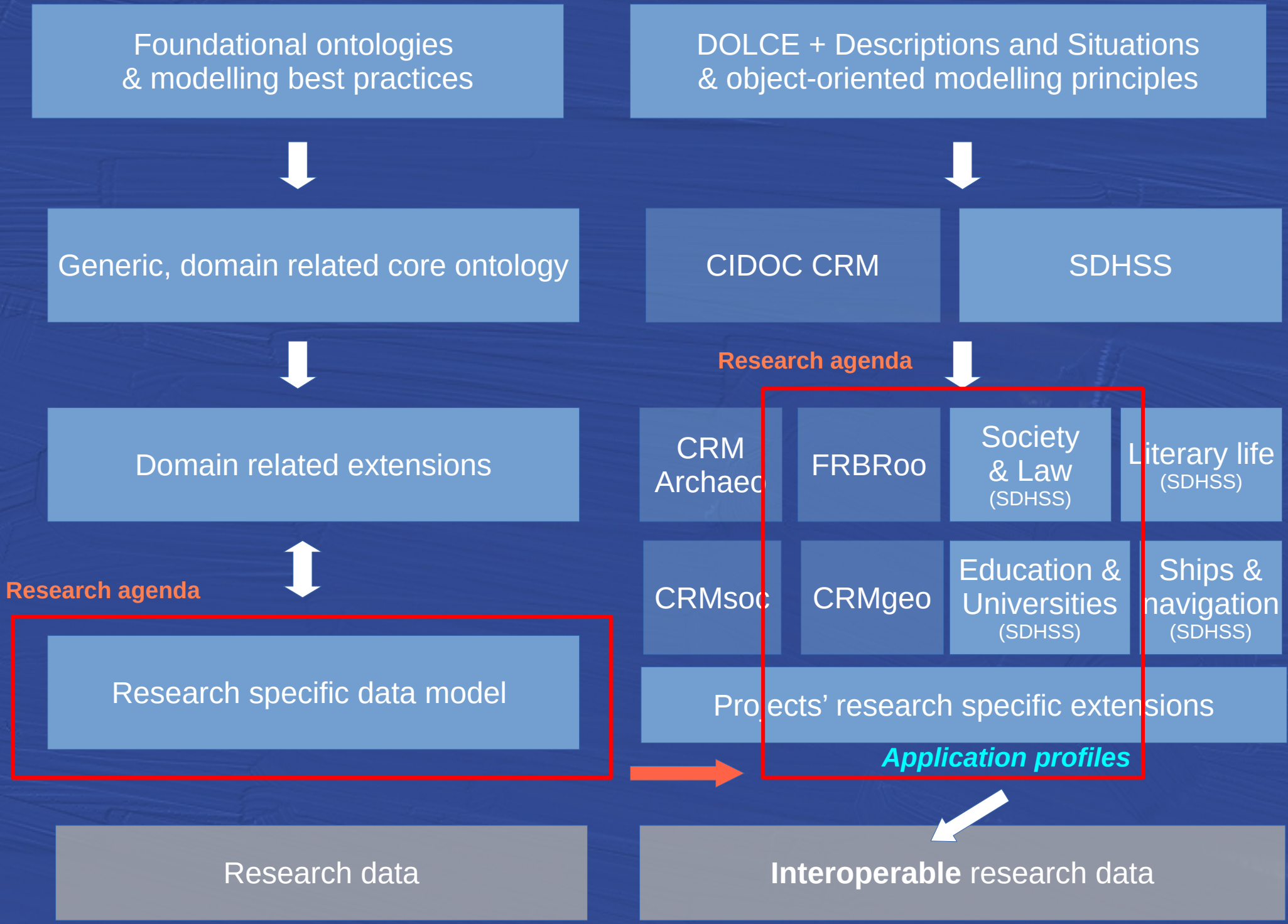
SDHSS

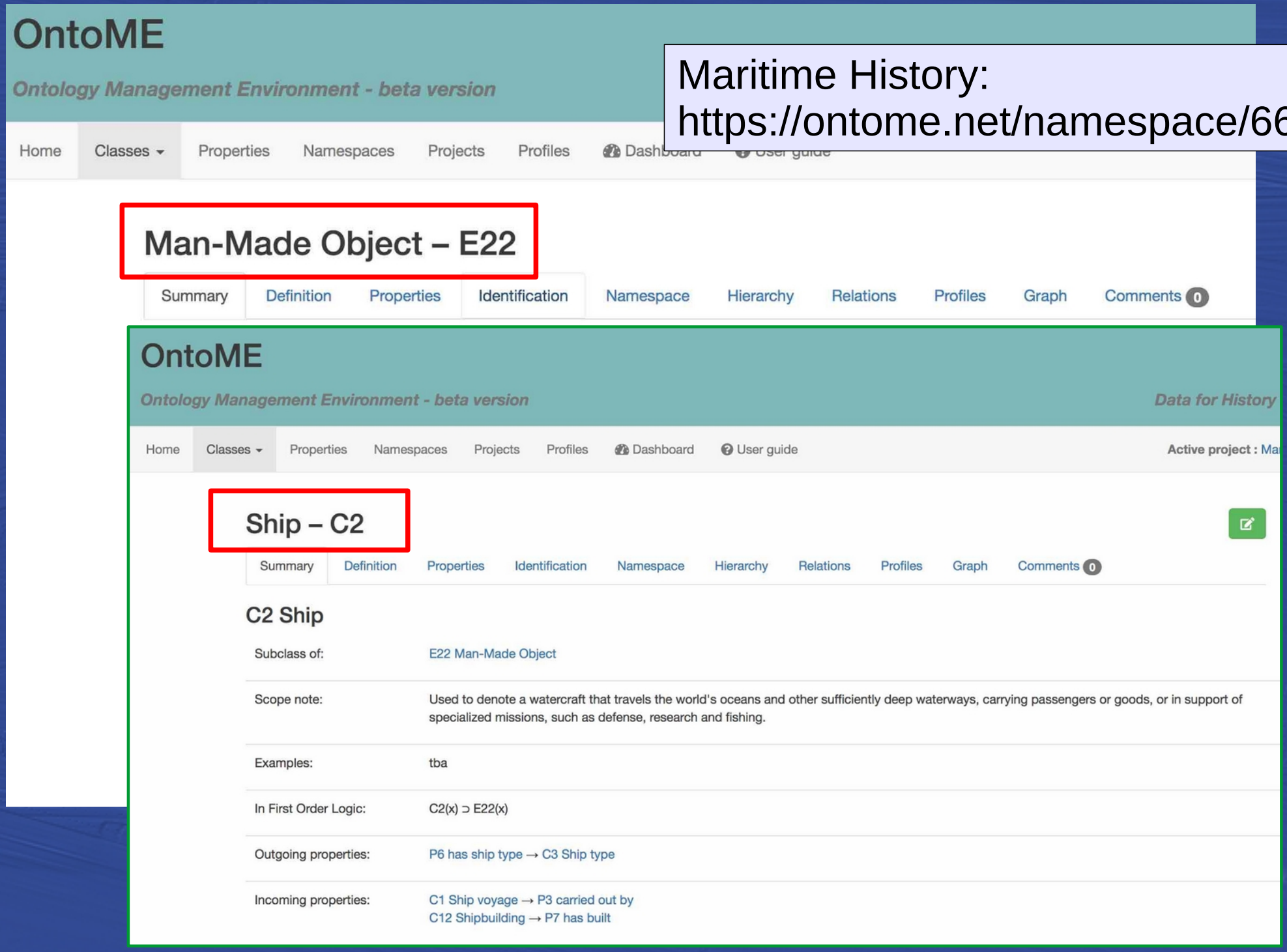
Research agenda

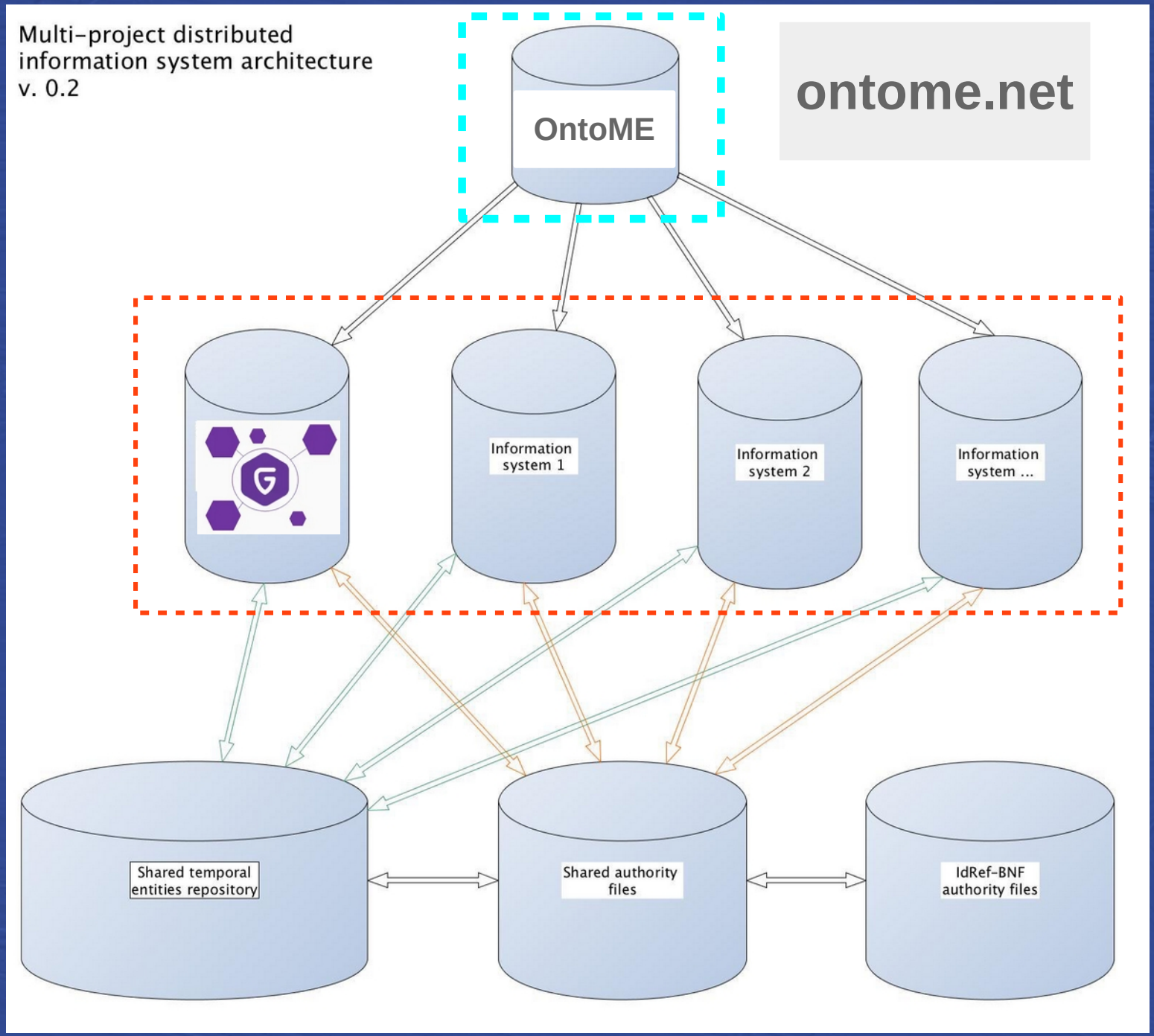
Research specific data model

Research data


Foundational ontologies
& modelling best practices
DOLCE + Descriptions and Situations
& object-oriented modelling principles
Generic, domain related core ontology
CIDOC CRM
SDHSS
Domain related extensions
CRM Archaeo
FRBRoo
Society & Law (SDHSS)
Literary life (SDHSS)
CRMsoc
CRMgeo
Education & Universities (SDHSS)
Ships & navigation (SDHSS)
Research agenda
Research specific data model
Research data


Foundational ontologies
& modelling best practices
DOLCE + Descriptions and Situations
& object-oriented modelling principles
Generic, domain related core ontology
CIDOC CRM
SDHSS
Research agenda
Domain related extensions
CRM Archaeo
FRBRoo
Society & Law (SDHSS)
Literary life (SDHSS)
Research agenda
CRMsoc
CRMgeo
Education & Universities (SDHSS)
Ships & navigation (SDHSS)
Research specific data model
Projects' research specific extensions
Application profiles
Research data
Interoperable research data


OntoME
Ontology Management Environment - beta version
Maritime History:
https://ontome.net/namespace/66
Home
Classes
Properties
Namespaces
Projects
Profiles
Man-Made Object – E22
Summary
Definition
Properties
Identification
Namespace
Hierarchy
Relations
Profiles
Graph
Comments 0
OntoME
Ontology Management Environment - beta version
Data for History
Home
Classes
Properties
Namespaces
Projects
Profiles
Dashboard
User guide
Active project :
Ship – C2
Summary
Definition
Properties
Identification
Namespace
Hierarchy
Relations
Profiles
Graph
Comments 0
C2 Ship
Subclass of:
E22 Man-Made Object
Scope note:
Used to denote a watercraft that travels the world's oceans and other sufficiently deep waterways, carrying passengers or goods, or in support of specialized missions, such as defense, research and fishing.
Examples:
tba
In First Order Logic:
C2(x) ⊃ E22(x)
Outgoing properties:
P6 has ship type → C3 Ship type
Incoming properties:
C1 Ship voyage → P3 carried out by
C12 Shipbuilding → P7 has built


Multi-project distributed
information system architecture
v. 0.2
OntoME
ontome.net
Information
system 1
Information
system 2
Information
system ...
Shared temporal
entities repository
Shared authority
files
IdRef-BNF
authority files

Some examples of projects making their data available in the Semantic Web



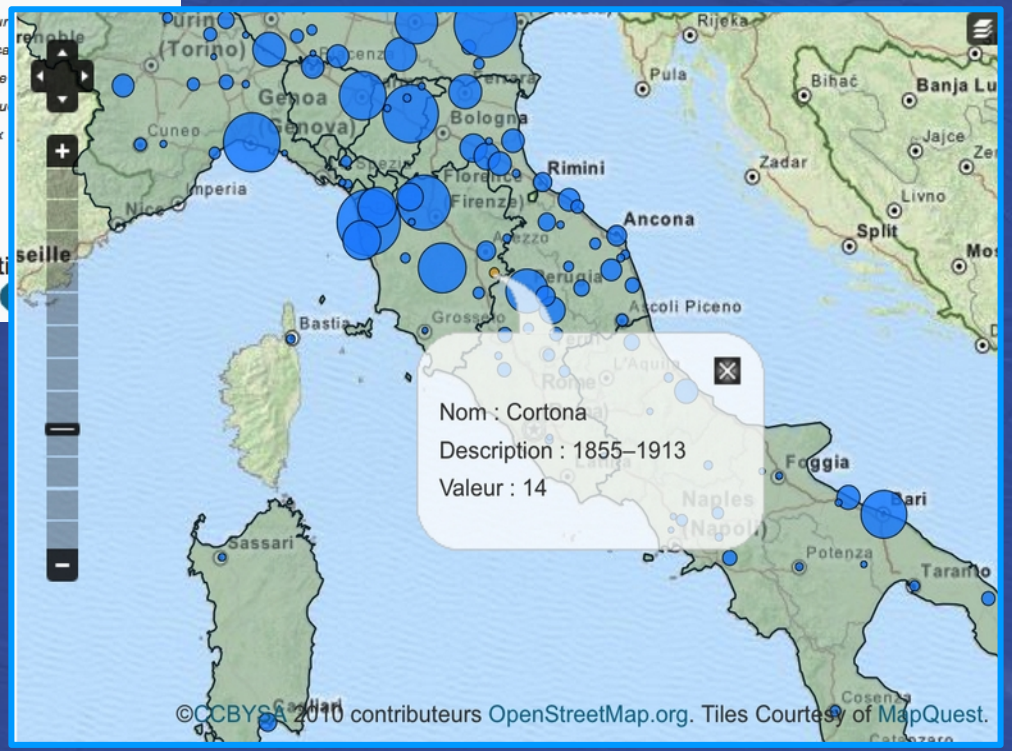
SPARQL – endpoint B3Kat

Bayerische Staatsbibliothek, Bibliotheksverbund Bayern, Kooperative Bibliotheksverbund Berlin-Brandenburg

https://lod.b3kat.de/doc/sparql-endpoint

Using library metadata published on the Semantic Web for historical research

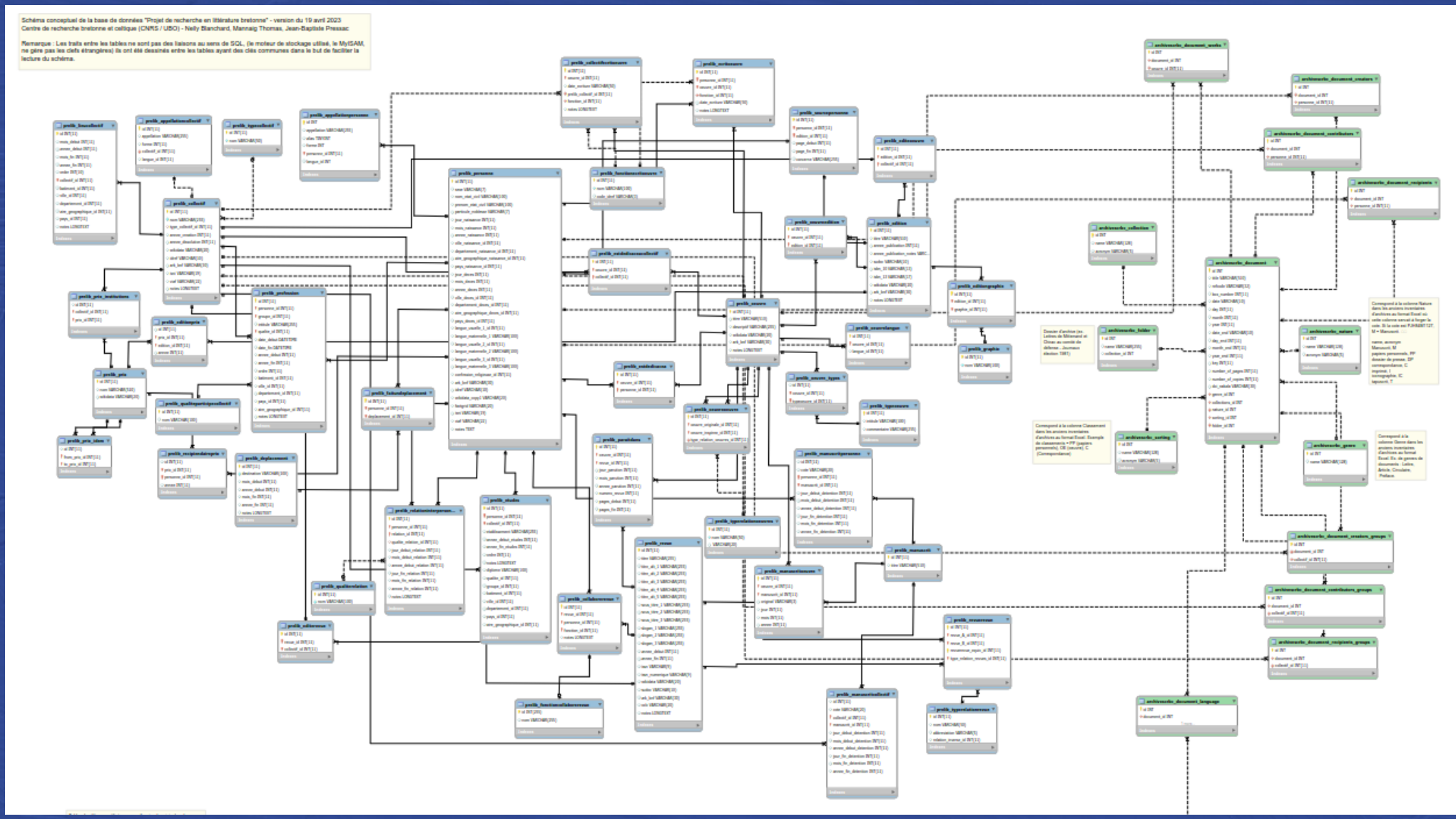




PRELIB

Projet de recherche en littérature de langue bretonne



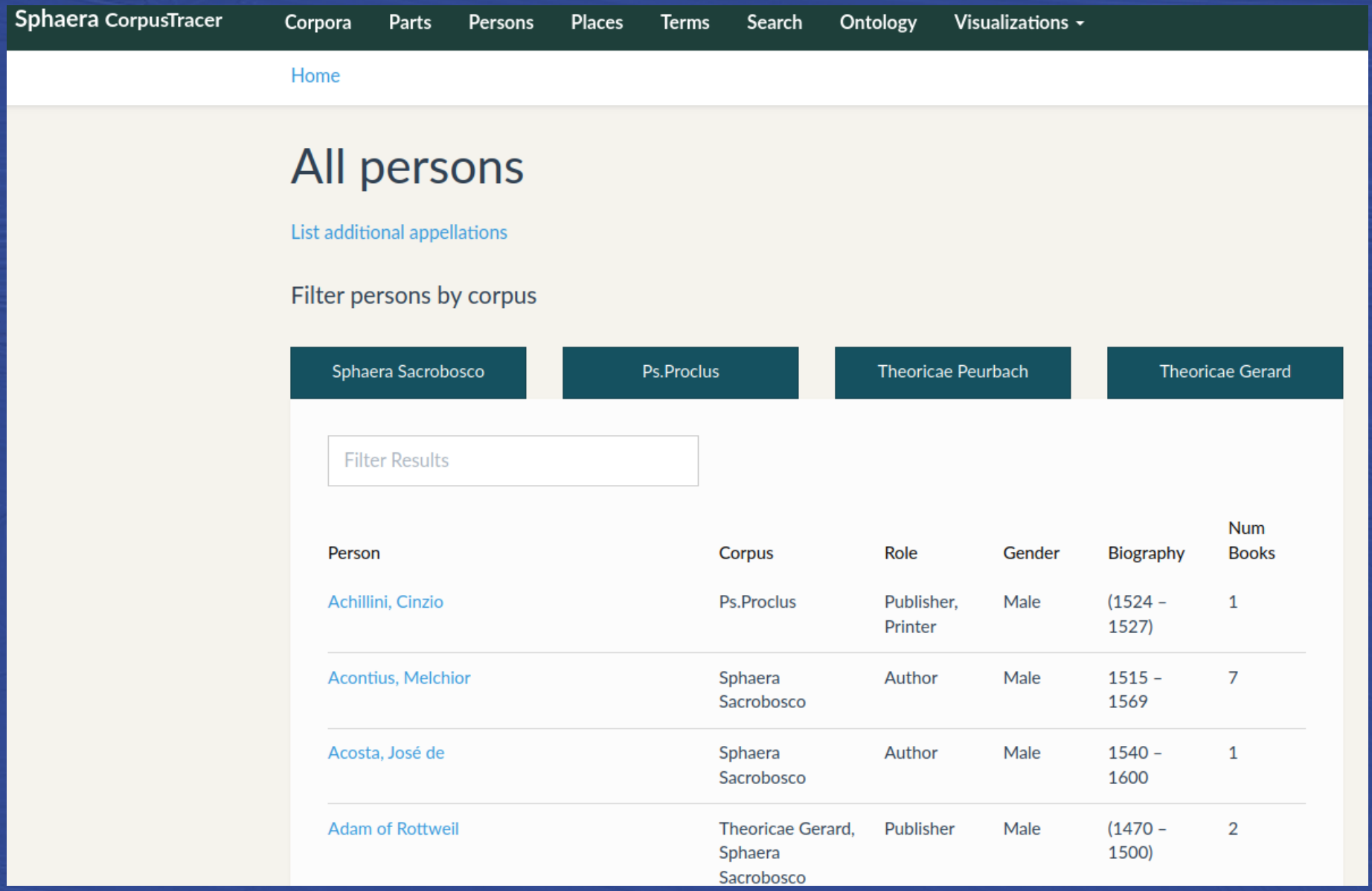

Schéma conceptuel de la base de données "Projet de recherche en littérature bretonne" - version du 19 avril 2023
Centre de recherche bretonne et celtique (CNRS / UBO) - Nelly Blanchard, Mannaig Thomas, Jean-Baptiste Pressac
Remarque : Les traits entre les tables ne sont pas des liaisons au sens de SQL, (le moteur de stockage utilisé, le MyISAM, ne gère pas les clefs étrangères) ils ont été dessinés entre les tables ayant des clés communes dans le but de faciliter la lecture du schéma.

Semantic Data for Humanities

Project for the publication of semantified data by ABES and LARHRA

https://dataforhumanities.org/sparql-endpoint/prelib-v1



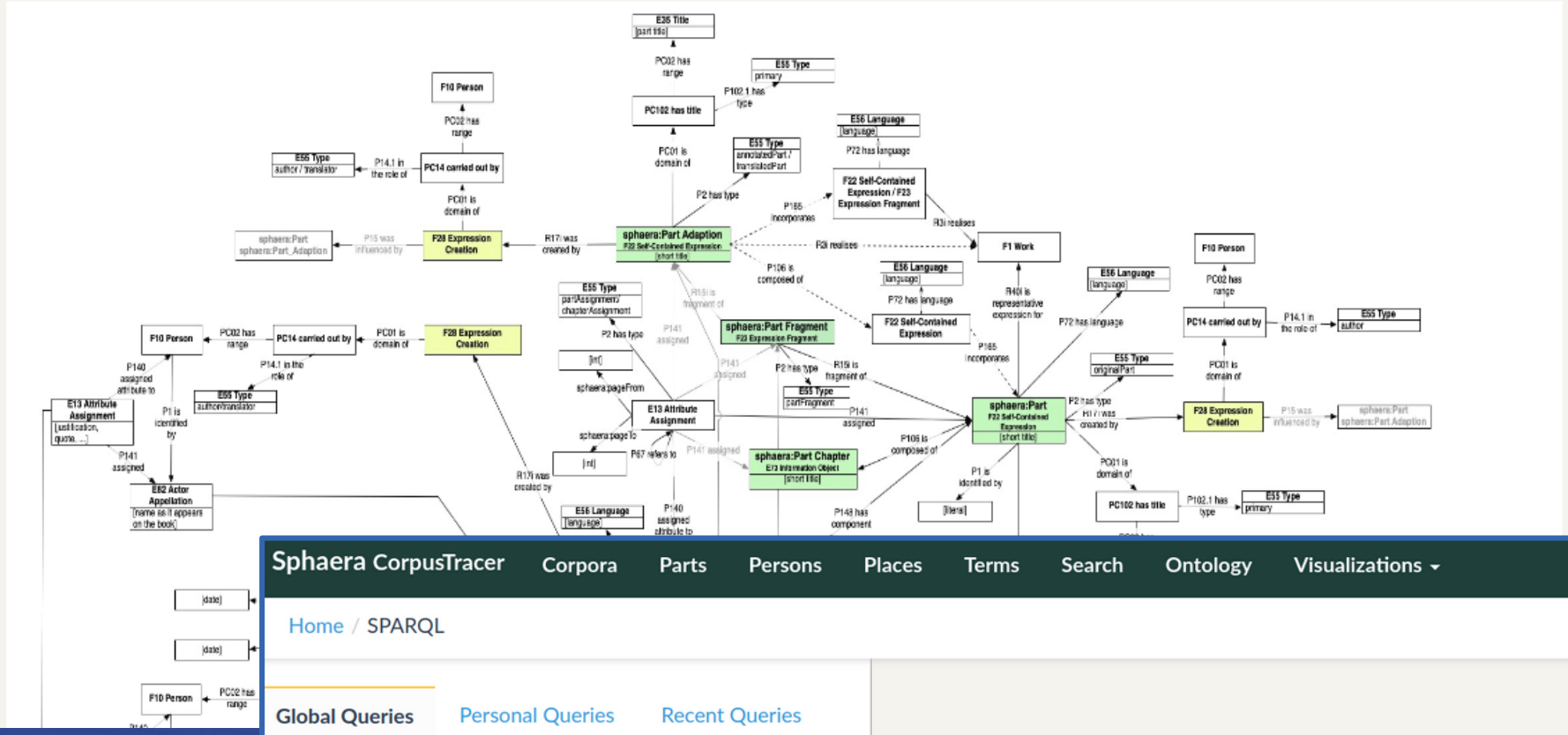


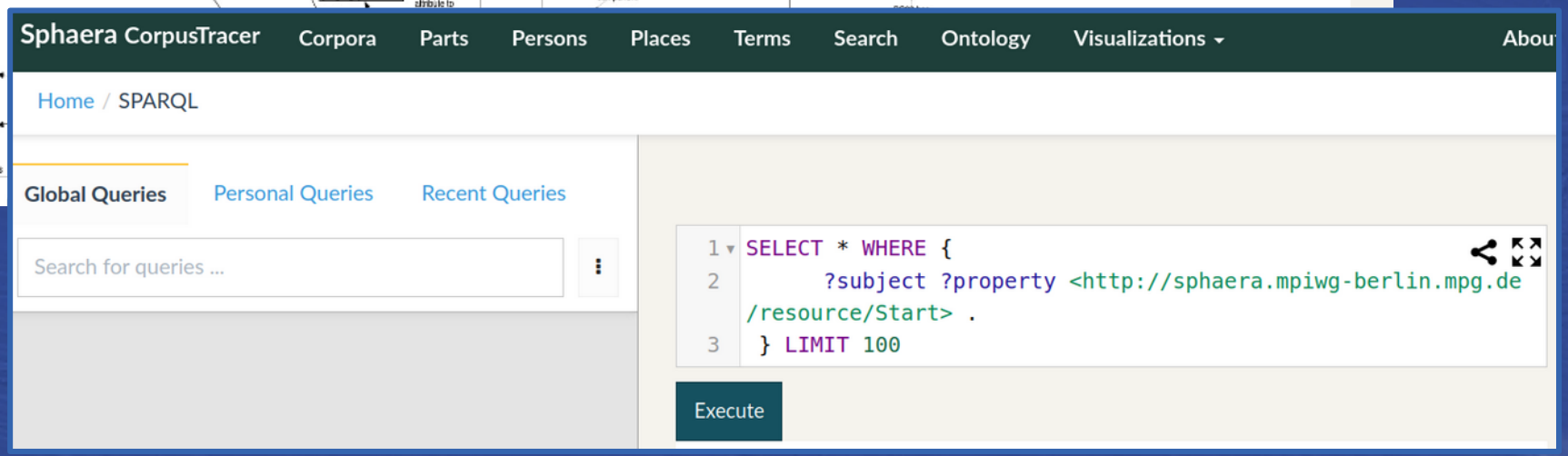
Kräutli, Florian, and Matteo Valleriani, 'CorpusTracer: A CIDOC Database for Tracing Knowledge Networks', Digital Scholarship in the Humanities, 33.2 (2018), 336–46 <https://doi.org/10.1093/llc/fqx047>

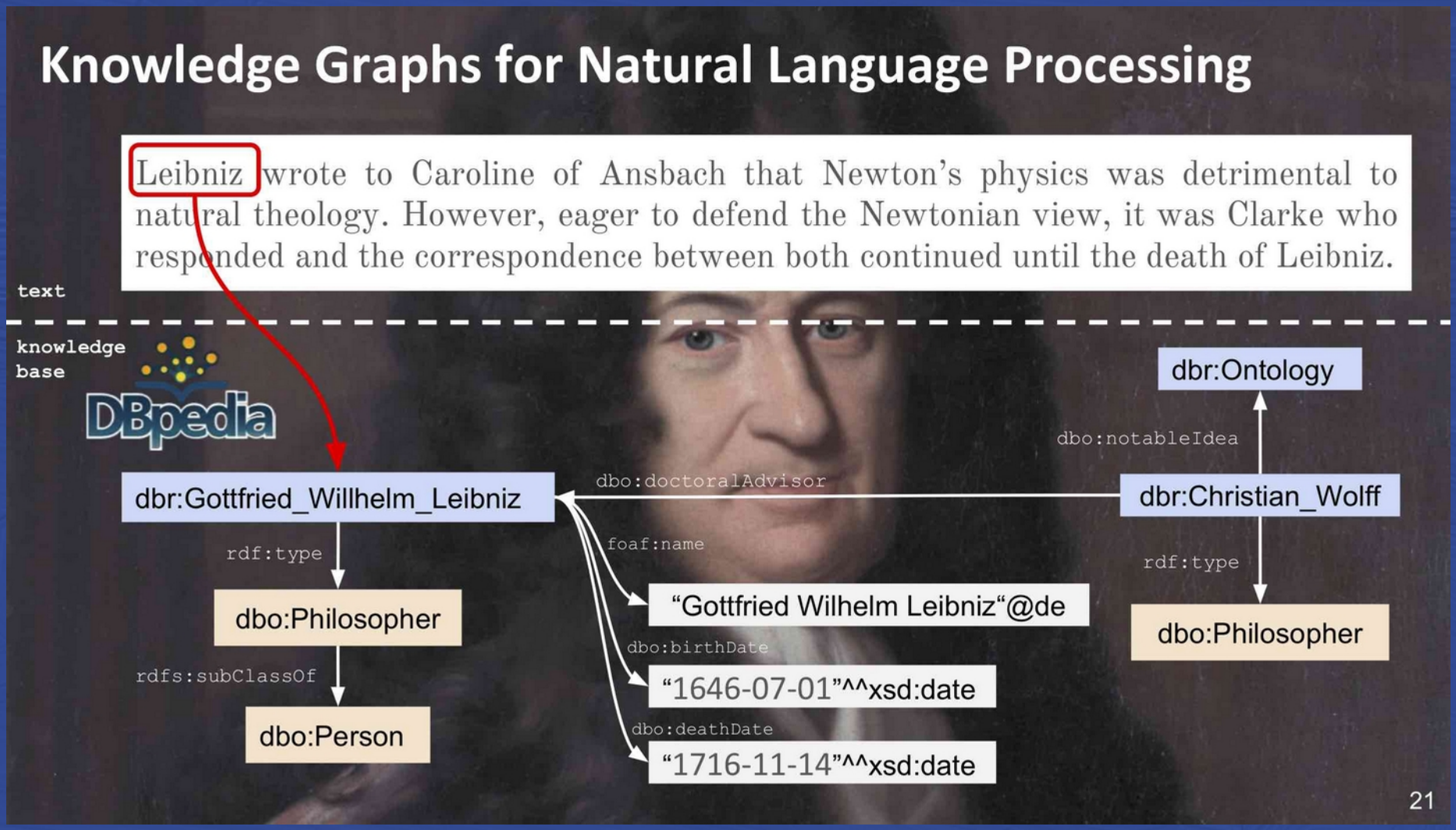
Sphaera Corpus Ontology Diagram

We model our data according to the formal ontology CIDOC-CRM and the FRBRoo extension for bibliographic records. This ensures that our data is compatible to other sources and remains usable in the foreseeable future. We use and extend the Erlangen OWL implementation of CIDOC-CRM and FRBRoo.

Book Relations







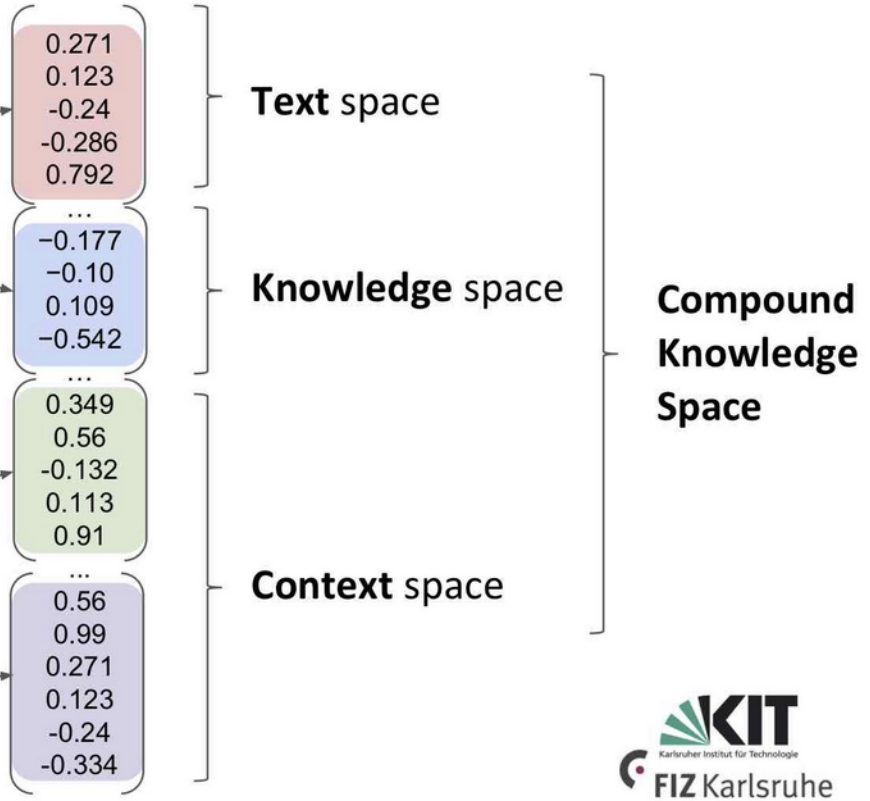
Karlsruher Institut für Technologie (29. Novembre 2017) – Antrittsvorlesung von

Prof. Dr. Harald Sack

Combining Semantics and Deep Learning for Intelligent Information Services

Combined Feature Embeddings for a Compound Knowledge Space

- Various feature vectors
  - **Word** embeddings
  - **Knowledge Graph** embeddings
    - Instances
    - Ontologies
  - Embeddings for **semantically enriched texts**
  - **Metadata** and **aggregated features**

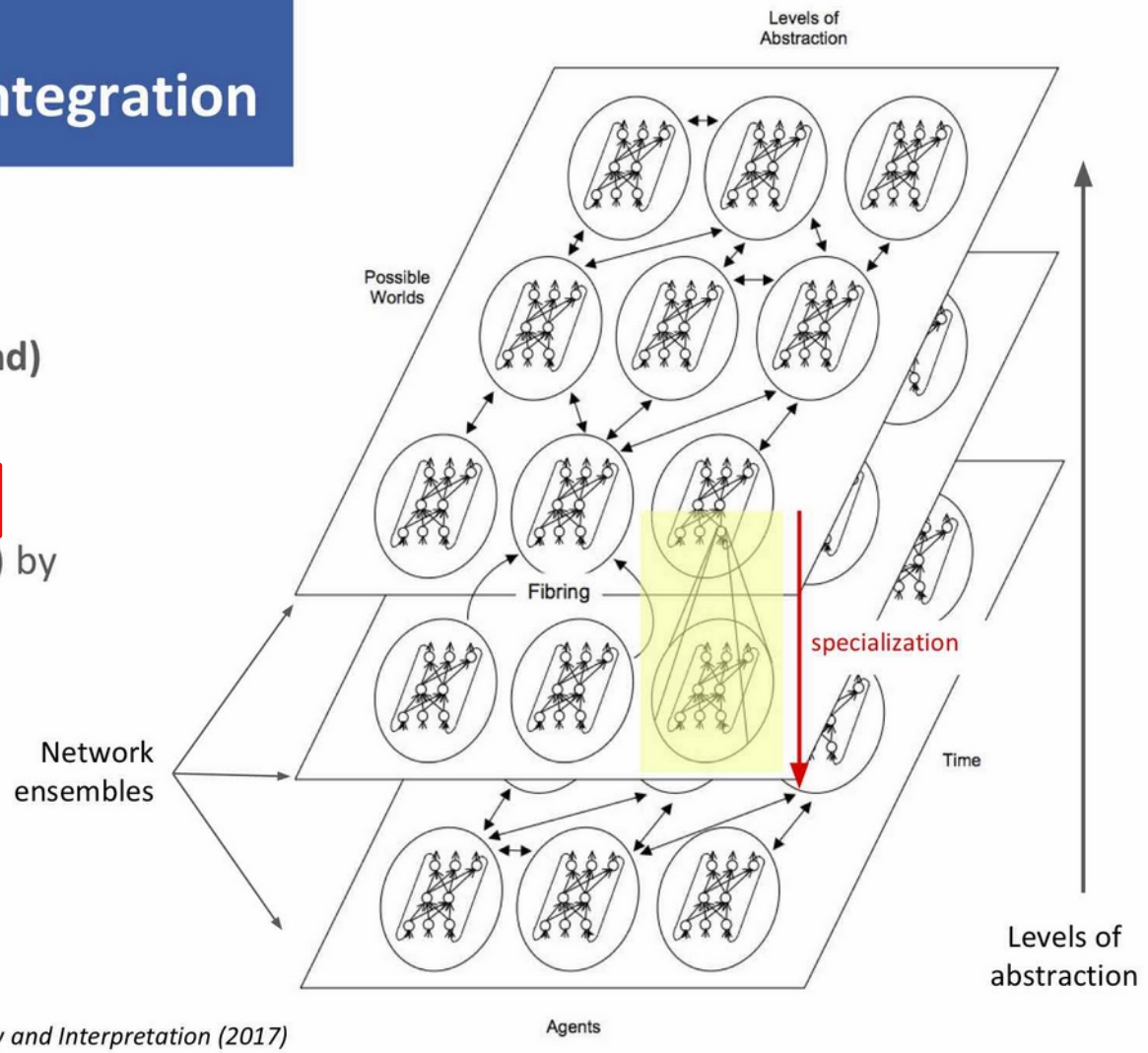


Antrittsvorlesung von Prof. Dr. Harald Sack (29. Novembre 2017)

Towards Neuro-Symbolic Integration

Neuro-Symbolic Systems

1. Translation of **symbolic (background) knowledge into the network**
2. **Learning of additional knowledge** from examples (and generalisation) by the network
3. **Executing the network** (i.e. reasoning), and
4. **Symbolic knowledge extraction** from the network.



33 *Besold et al.: Neural-Symbolic Learning and Reasoning: A Survey and Interpretation (2017)*

Antrittsvorlesung von Prof. Dr. Harald Sack (29. Novembre 2017)


LOD4HSS
Linked Open Data for the
Humanities & Social Sciences

Thank you !